Introduction:

There are discoveries and inventions which change the course of civilization. Invention of semi-conductor devices is one such event which to this day is bringing about continuous evolution of technology. Computers, mobile phones, televisions and all electronic devices are based on applications of small scale circuit elements like diodes and transistors. Thus semi-conductors are one of the most important dimensions of technologies shaping up the direction of the world and the civilization. In this chapter we will be learning about various types of semi-conductor devices, such as diodes, transistors, logic-gates and their applications.

Classification Of Metals, Semi-conductors & Insulators Based On Conductivity:

On the basis of the relative values of electrical conductivity (σ) or resistivity ($\rho = 1/\sigma$), the solids are broadly classified as:

(i) ***Metals:*** They possess very low resistivity (or high conductivity).
$\rho \sim 10^{-2} - 10^{-8}\ \Omega\ m$
$\sigma \sim 10^{2} - 10^{8}\ S\ m^{-1}$

(ii) ***Semiconductors:*** They have resistivity or conductivity intermediate to metals and insulators.
$\rho \sim 10^{-5} - 10^{6}\ \Omega\ m$
$\sigma \sim 10^{5} - 10^{-6}\ S\ m^{-1}$

(iii) ***Insulators:*** They have high resistivity (or low conductivity).
$\rho \sim 10^{11} - 10^{19}\ \Omega\ m$
$\sigma \sim 10^{-11} - 10^{-19}\ S\ m^{-1}$

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The values mentioned in the above table are only indicative. Infact, resistivity is not the only criterion for distinguishing between a metal and an insulator or a semi-conductor. There exists a more sophisticated theory known as band theory of solids which gives us deeper insight in to material behaviour with regard to their electrical behavior. So far just based on resistivities, it still remains a mystery as to what is so special about semi-conductors. Infact the answers to the mystery lies in the electronic configuration, interaction of atoms in the crystal and quantum mechanics about which we will do some unraveling in

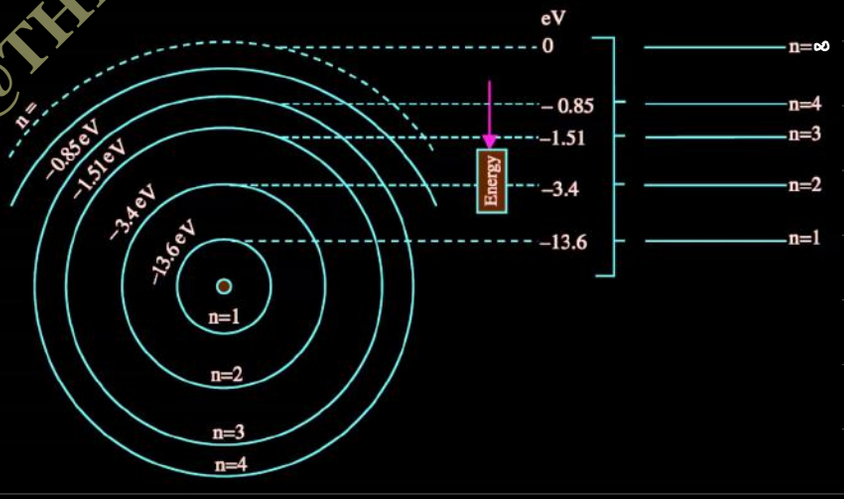
the upcoming sections.

Types Of Materials Used As Semi-Conductors:

(i) *Elemental semiconductors*: Si and Ge
(ii) *Compound semiconductors*: Examples are:
- Inorganic: CdS, GaAs, CdSe, InP, etc.
- Organic: anthracene, doped pthalocyanines, etc.
- Organic polymers: polypyrrole, polyaniline, polythiophene, etc.

Most of the currently available semi-conductors are based on elemental semiconductors Ge and Si and compound inorganic semi-conductors however since 1990s organic and polymer semi-conductors have been built which will pave the way for futuristic technologies of polymer electronics and molecular-electronics, however in our discussion at JEE level we will restrict our discussion mostly to Si and Ge.

Formation Of Energy Bands In Solids:



Recall that permissible energy levels in an atom are governed by quantum mechanics, i.e schrodinger equation and the potential function defined in space. With discrete atoms far away from each other, we have a very simple central potential field leading to discrete energy levels such as the ones in a hydrogen atom as shown in the figure above. However when atoms come together in form of a crystal, they experience the electromagnetic fields from neighboring atoms also and accordingly instead of unique sharp energy levels, we start getting nearly continuous bands of energies instead of sharp unique energy levels. This splitting of energy levels gets more and more pronounced as the atoms come nearer and nearer. (This is because as the lattice spacing decreases, the effect of electromagnetic field of

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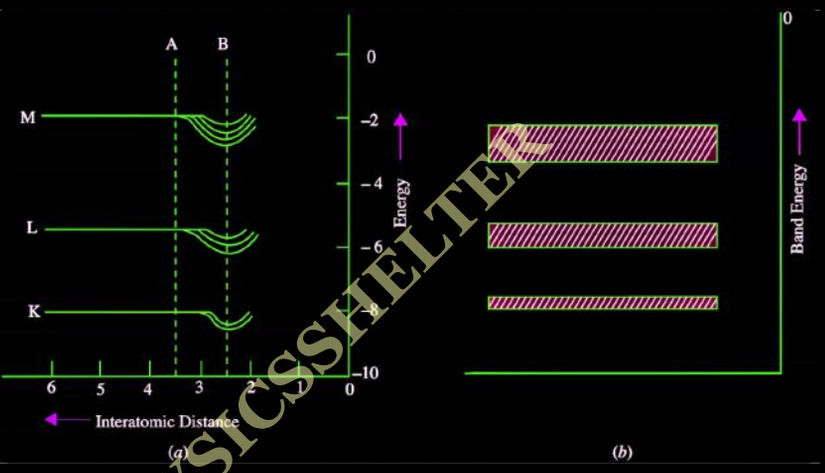
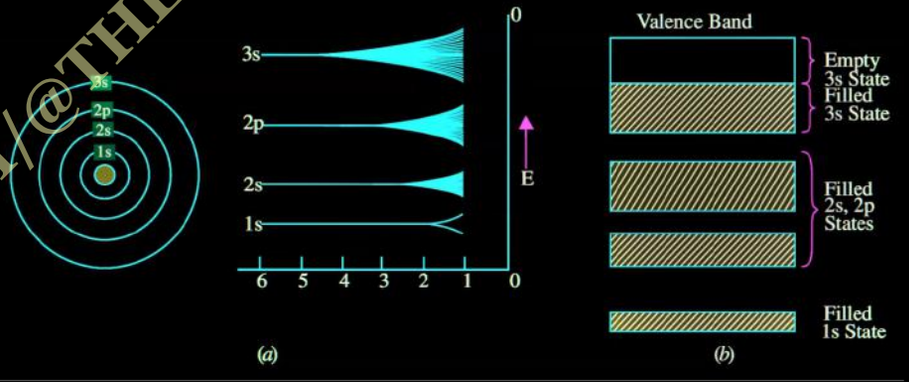
neighboring atoms becomes stronger and stronger thus increasing the distortion in the net field (as compared to the simple central field created by a single nucleus). If there are N number of atoms in the crystal, each energy level splits in to N closely spaced energy levels (somewhat similar to what you learn about bonding and anti-bonding orbitals in molecular orbital theory in chemistry).

As an example, adjacent figure schematically shows splitting of K, L and M shells as the spacing between atoms is reduced. At first only valence level or M level is affected (marked by vertical dotted line A) and as spacing is reduced, inner shells also become affected (see dotted line B).



As another example consider the adjacent figure describing splitting of energy levels in a sodium crystal. The configuration of a sodium atom at 0 Kelvin is $1s^2 2s^2 2p^6 3s^1$. Here 3S subshell is only half filled can it can occupy one more electron. Accordingly when this subshell splits in to N levels (∵ there are N atoms), half of these are filled and other half is empty at absolute zero. (Although with higher temperatures electrons can jump to higher levels from lower levels).



VALENCE BAND: The highest energy band in a crystal which is either fully or partially filled is called the valence band (This is the band filled by valence electrons). (We are talking about filling at 0 Kelvin).

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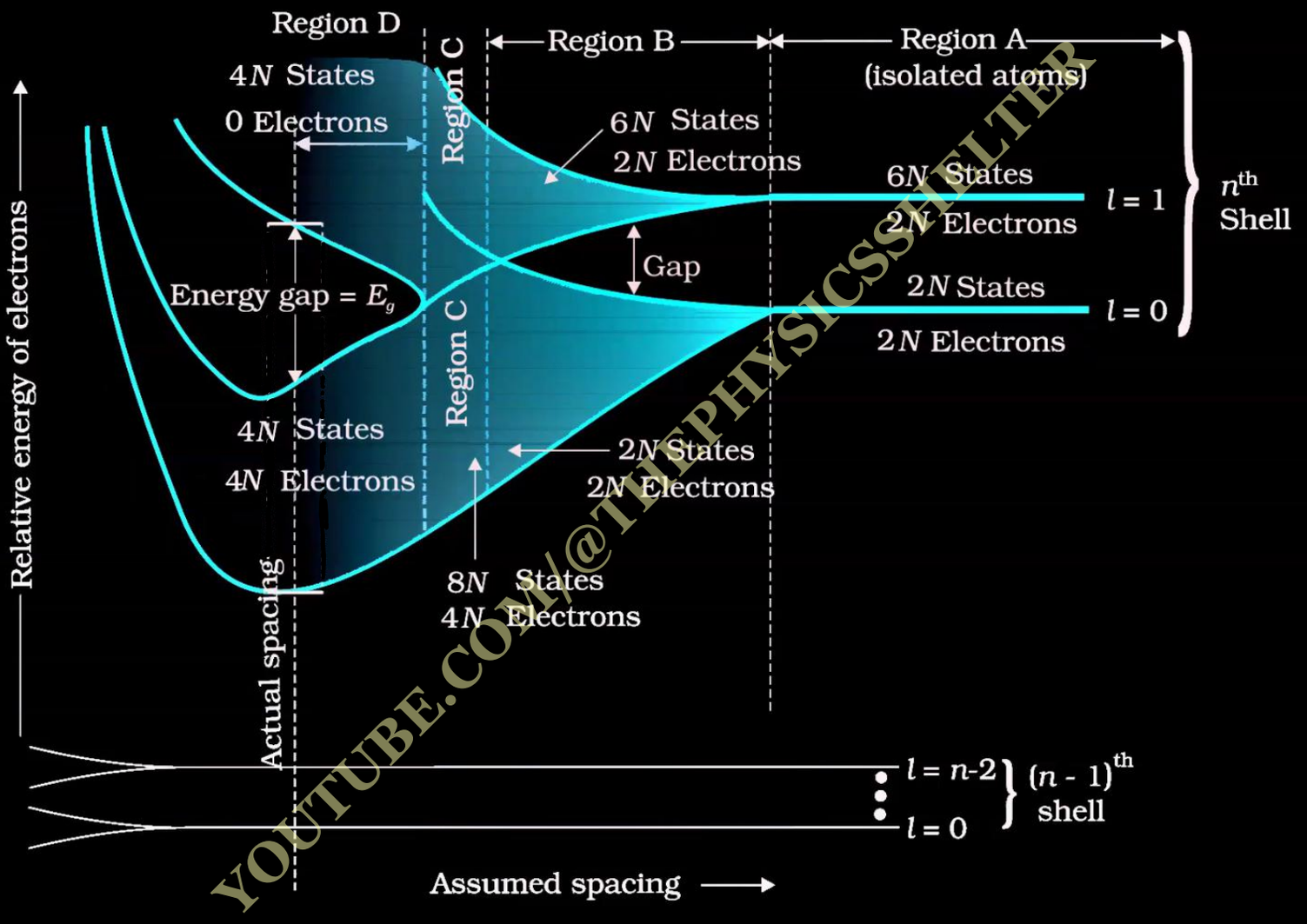
CONDUCTION BAND: The lowest energy band which is partially vacant or fully vacant (if there are no partially vacant bands) at 0 Kelvin is called conduction band. In case of partially vacant band, the conduction band is said to overlap with valence band.

BAND STRUCTURE IN A SILICON CRYSTAL:



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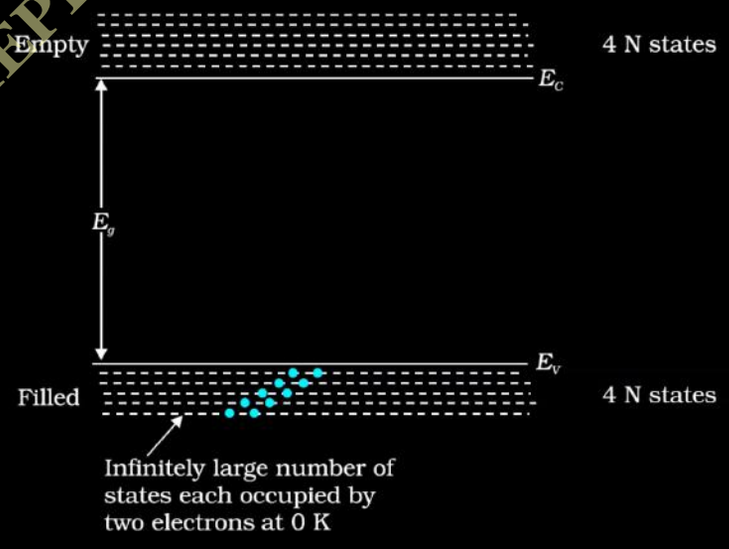
When a silicon atom is isolated, its electronic configuration is given as $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$. Thus if we consider N isolated atoms, We have 6N energy states corresponding to 3p out of which 2N are occupied and another 2N states corresponding to 3s, both of which are occupied (with 2N electrons). This situation is represented by the rightmost region in the above graph. Now when the atoms are further brought closer together, the energy band splitting of the levels starts. Both 6N states corresponding to 3p orbitals and 2N states

corresponding to 3s states form wider and wider bands as the separation reduces, however still there is a gap between the bands corresponding to splitting of 3p and 3s energy levels. This is known as band gap and represented in region B in the figure. At the left end of the region B, the band gap becomes zero. Now when the separation between atoms is further reduced, we have a region of continuous band of 8N energy states out of which 4N are filled and another 4N are vacant. This corresponds to the region C in the figure. At the left end of region C, overlapping stops and band gap appears once again as lattice spacing is further decreased (region D). The dotted line in the region D represents actual spacing of the silicon lattice. (If lattice constant were further decreased, it would increase the band gap and decrease the width of valence band).

WHY IS SILICON A POOR CONDUCTOR AT ZERO KELVIN:

For conduction to be possible, electrons of the material should be able to jump from one energy state to another energy state. However at absolute zero, the valence band of silicon is completely occupied and electrons have no vacant states where they can jump. In conduction band there are only vacancies but no electrons to jump. So conduction is not possible unless some electrons are made available in the conduction band or some vacancies are created in the valence band. (You can think of situation like simultaneous presence of a Highway without any bikes and a narrow lane with too many bikes stuck in a traffic Jam. If some bikes could shift to the highway, traffic would be possible in both places. This happens due to thermal excitation when the temperature is increased).



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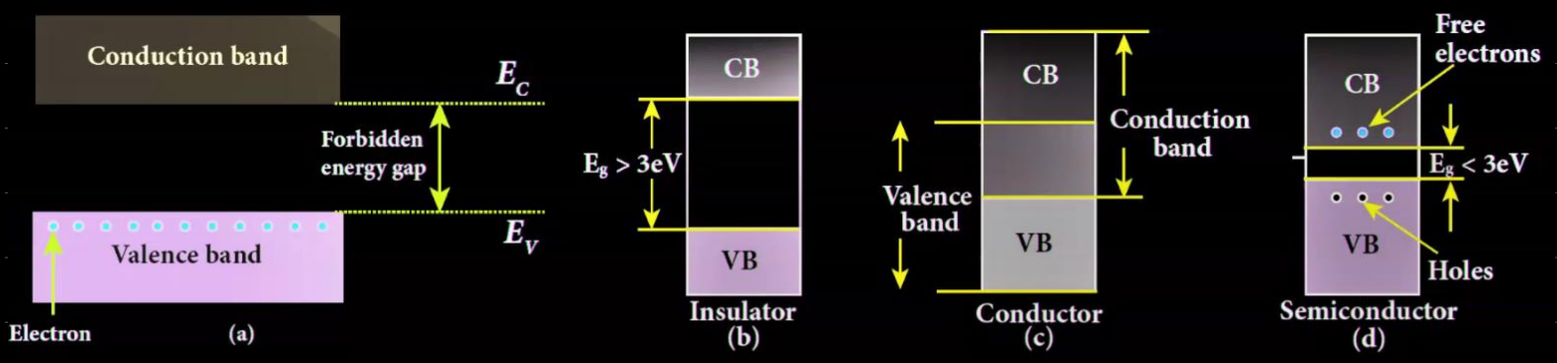
Classification Of Materials Based On Band Theory:



(a) Schematic representation of valence band, conduction band, and forbidden energy gap. Energy band structure of (b) Insulator (c) Conductor (d) Semiconductor

Insulators: For these materials, band gap is more than 3eV (typically mostly 6eV or more) accordingly at room temperature, electrons are not able to jump from valence band to conduction band. (At room temperature, $kT \simeq 1/40$ eV). Thus the conduction band practically remains vacant while valence band without any vacancies for movement of electrons.

Semiconductors: Here the band gap is less than 3eV (for silicon it is 1.1 eV and for Ge it is 0.7 eV). When temperature of the crystal rises some electrons jump from valence band to conduction band thus making conduction possible upon application of EMF across the crystal.

Conductors: Conductors have overlapping conduction and valence bands, thus there is no band gap. Due to partially filled continuous band of energy, electrons can freely move from one energy state to another (even at absolute zero).

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INTRINSIC SEMI-CONDUCTORS:

A pure crystal of a semi-conductor material such as Si or Ge is called an intrinsic semi-conductor crystal. At low temperatures all the electrons are locked in the form of covalent bonds with adjacent atoms. However as the temperature rises, more thermal energy becomes available and some of the electrons may break free from covalent bonds and start participating in conduction.

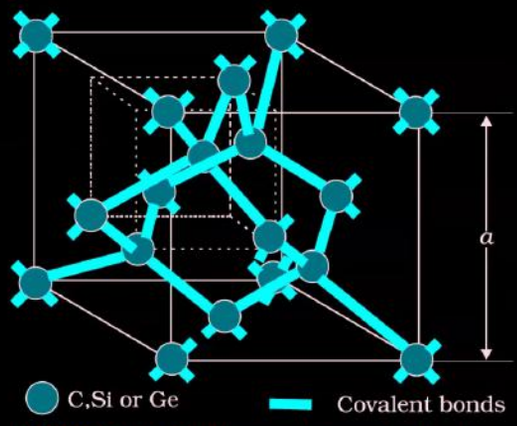
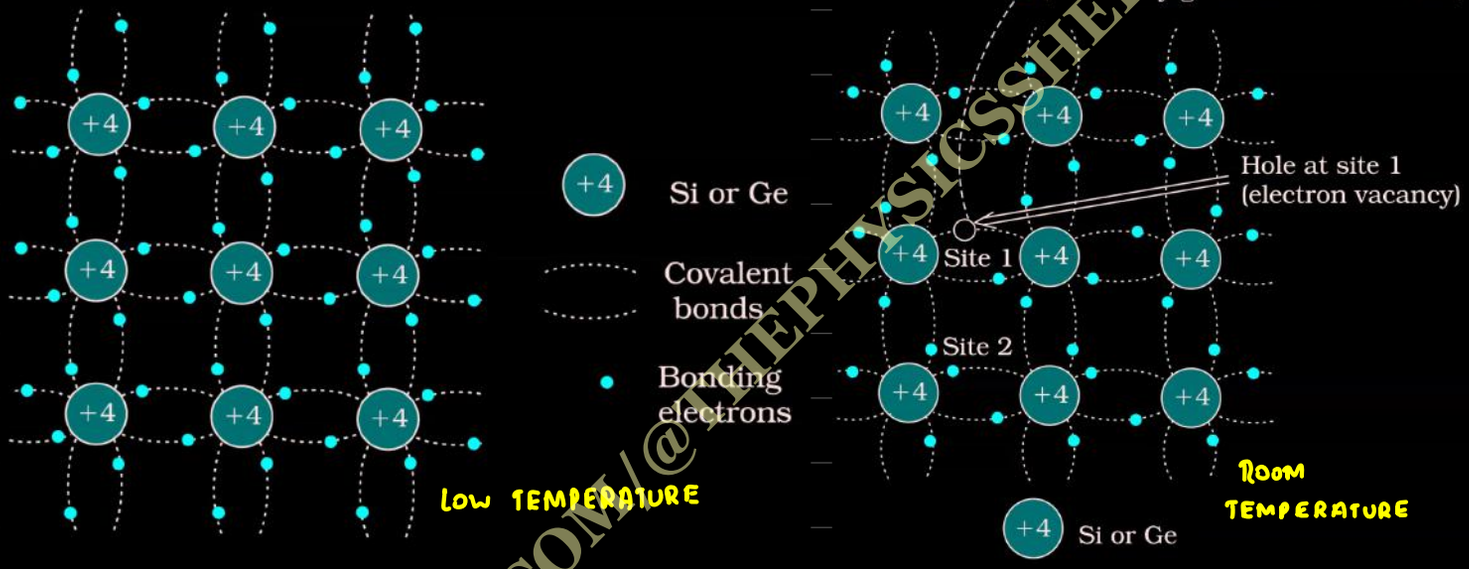
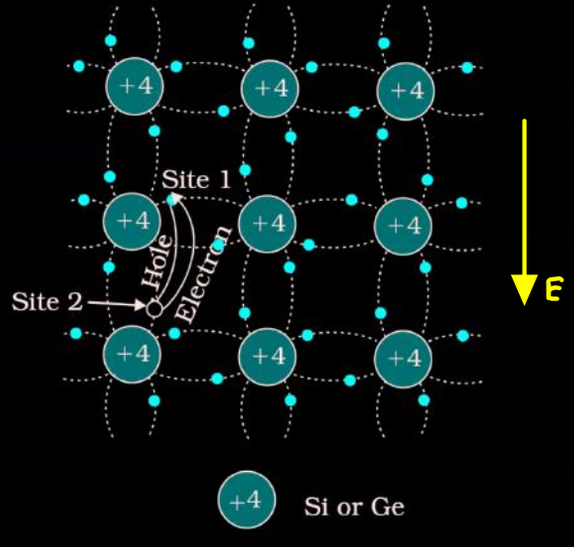


FIGURE 14.3 Three-dimensional diamond-like crystal structure for Carbon, Silicon or Germanium with respective lattice spacing a equal to 3.56, 5.43 and 5.66 Å.



When an electron breaks free from covalent bond, a vacancy is created there (atom becomes an ion), such a vacancy is called a hole and upon application of electric field, it may be systematically filled by electrons elsewhere in the valence bonds, thus conduction is made possible. When such a crystal is exposed to electric field, electrons have a tendency to move against electric field. If a valence electron jumps from one covalent bond to another, it leaves a vacancy in its original position. Thus it looks as if the hole has moved along the electric field. In that sense, holes are assumed to be like positive charges



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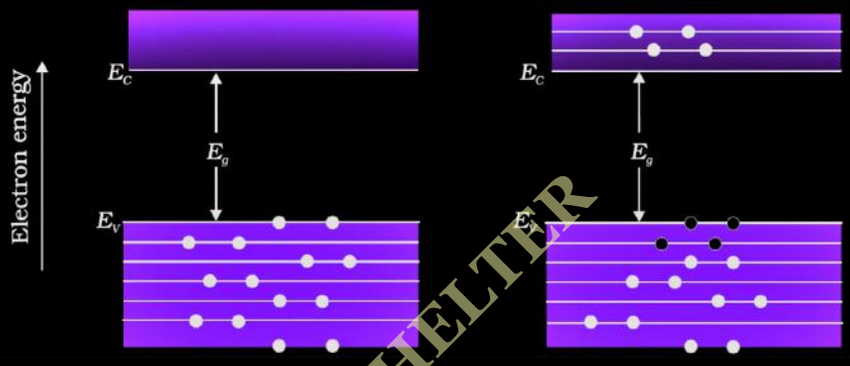
(charge +e) for the purpose of calculating electric current. The thermally generated electrons will also move against the applied field, thus contributing to the current (such electrons are in the conduction band). Net current is the summation of conduction current and the hole current.

Thermal Pair Generation:



As temperature of a crystal increases, more and more electrons jump from valence band in to conduction Band thus creating an equal number of holes in the intrinsic semi-conductor. The concentration of holes on electrons per unit volume is governed by quantum and statistical mechanics and is given as $n_i = N e^{-E_g/2kT}$ where N is a constant for a given semi-conductor, E_g is the band gap and T is the temperature. E_g for silicon is $3 \times 10^{25}/m^3$.

Fermi Level And Fermi-Dirac Statistics:

The probability of an electron occupying an energy state of energy E is given by fermi-dirac statistics i.e.

$$f(E) = \frac{1}{1 + e^{(E-E_f)/kT}}$$

Here E_f is called fermi-level and represents a hypothetical energy level whose probability of occupation is 50% (if not forbidden). At zero Kelvin all states below fermi level have 100% probability of being filled and all states above it will have 0% probability of being filled. Thus fermi level separates the filled and unfilled energy bands at 0K and hence lies between conduction band and valence band. The variation of fermi-dirac distribution function is given in the figure on following page. As the temperature increases, the states above fermi level have an increased

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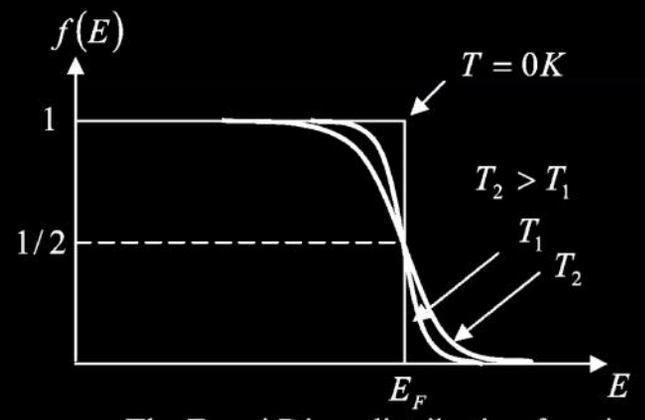
probability of filling (though still less than 50%) while the states below fermi level have a decreased probability of being filled (though still above 50%). The terms Fermi level and Fermi energy are used interchangeably though fermi-energy is defined as energy of the highest occupied level at 0K.



The Fermi Dirac distribution function.

For semiconductors, fermi-level is taken as the mid point of the band gap as shown in the figure. Thus Fermi level is the average of highest energy in the valence band and lowest energy of the conduction band. Fermi level divided by boltzmann constant is called Fermi temperature and represents a Temperature where thermal energies become comparable to the Fermi Energy.

Context of Fermi level for a semiconductor

EXTRINSIC SEMI-CONDUCTORS: Intrinsic semi-conductors have very low conductivity and are therefore not very useful directly. Therefore to improve their conductivity, some of the atoms are replaced by either Pentavalent impurities (like Phosphorus) or Trivalent impurities (like Aluminum). This process of replacing the intrinsic semi-conductor atoms is called doping. If doping is done with pentavalent impurities we call it N type doping while if we dope with trivalent impurities, we call it P type doping.

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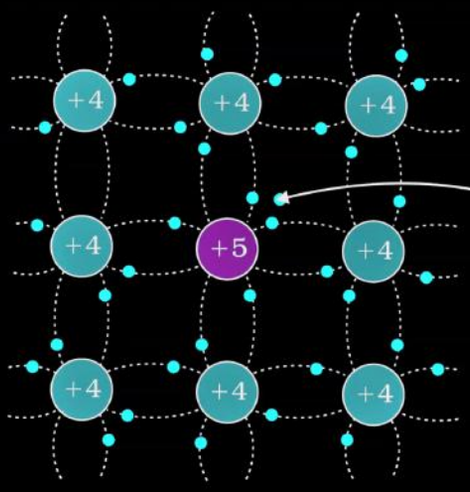
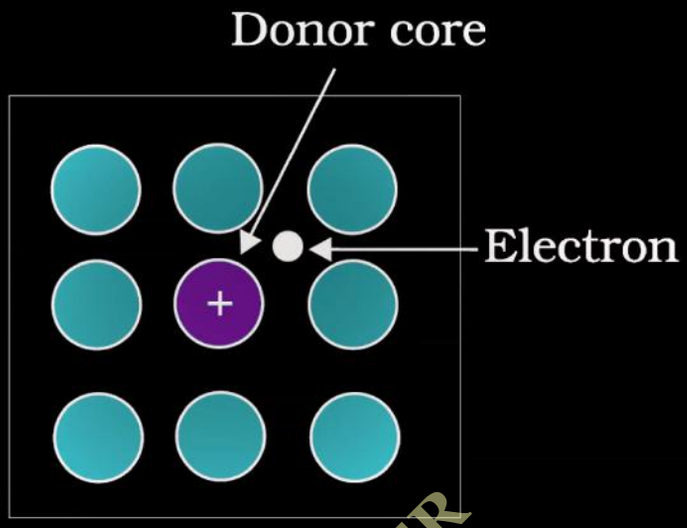
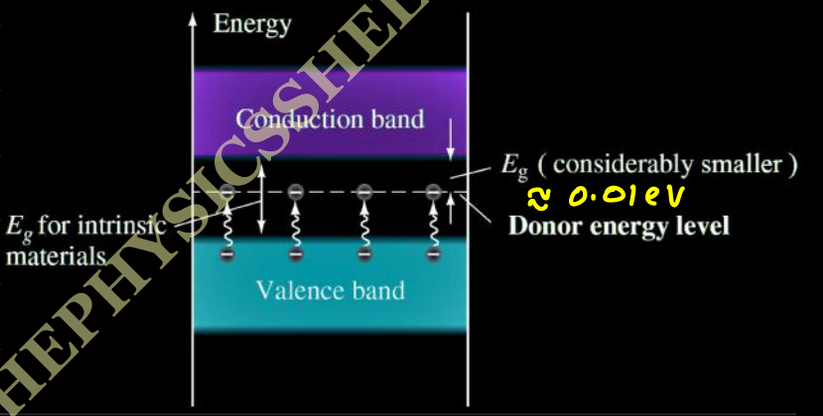
N Type Semi Conductors:





Pentavalent impurities are able to form four covalent bonds while their fifth electron easily passes in to the conduction band. This is because energy gap of the donor from the conduction band is very small as compared to that of intrinsic semiconductors (N type doping increases the fermi level). Since the donor atoms add an extra electron to the crystal, thus, N type crystals have more number of electrons than holes. (Infact increased number of electrons increase the hole-electron recombinations thus an N type crystal has even smaller number of holes than an intrinsic crystal). Following equation holds true for N type doped crystals:



$$n_e n_h = n_i^2$$

n_e: electron concentration n_h: hole concentration
n_i: electron or hole concentration in intrinsic crystal.

In a typical doping situation, practically the concentration of electrons is almost equal to concentration of N type dopant. For calculation of electrons and holes in a crystal we can do the following analysis. Let n_D be the donor dopant concentration and let x be the electron hole recombination per unit volume then,

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$n_e = n_i + N_D - x$ and $n_h = n_i - x$ thus:

$$\boxed{n_e - n_h = N_D} \quad (1) \quad \text{and} \quad \boxed{n_e n_h = n_i^2} \quad (2)$$

An exact solution of equations (1) & (2) would involve solving a quadratic but in most situations, numbers allow us to use approximations for easy calculations.

→ Since pentavalent impurity donates it atom it get ionized, thus the crystal will have ions of P^+ or Sb^+ types of atoms.

NCERT EXAMPLE:

Example 14.2 Suppose a pure Si crystal has 5×10^{28} atoms m^{-3}. It is doped by 1 ppm concentration of pentavalent As. Calculate the number of electrons and holes. Given that $n_i = 1.5 \times 10^{16}$ m^{-3}.

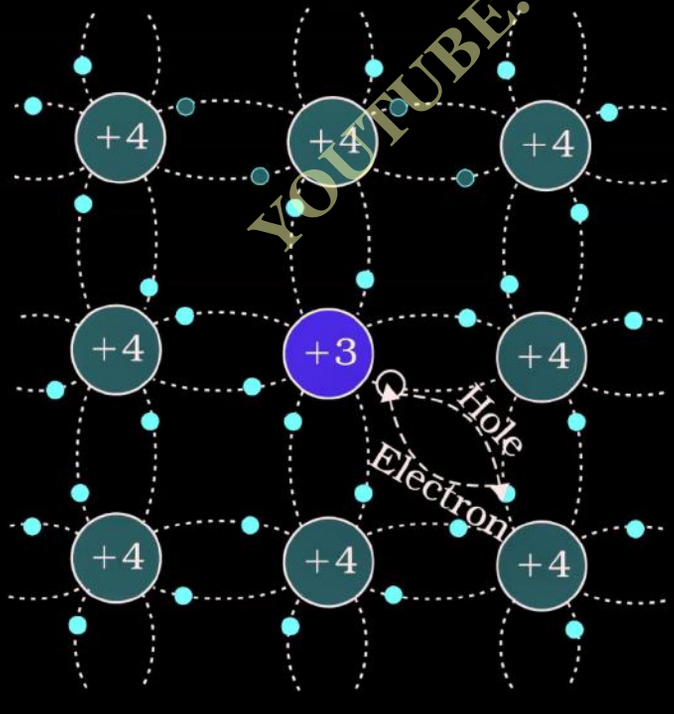
Solution Note that thermally generated electrons ($n_i \sim 10^{16}$ m^{-3}) are negligibly small as compared to those produced by doping.
Therefore, $n_e \approx N_D$.
Since $n_e n_h = n_i^2$, The number of holes

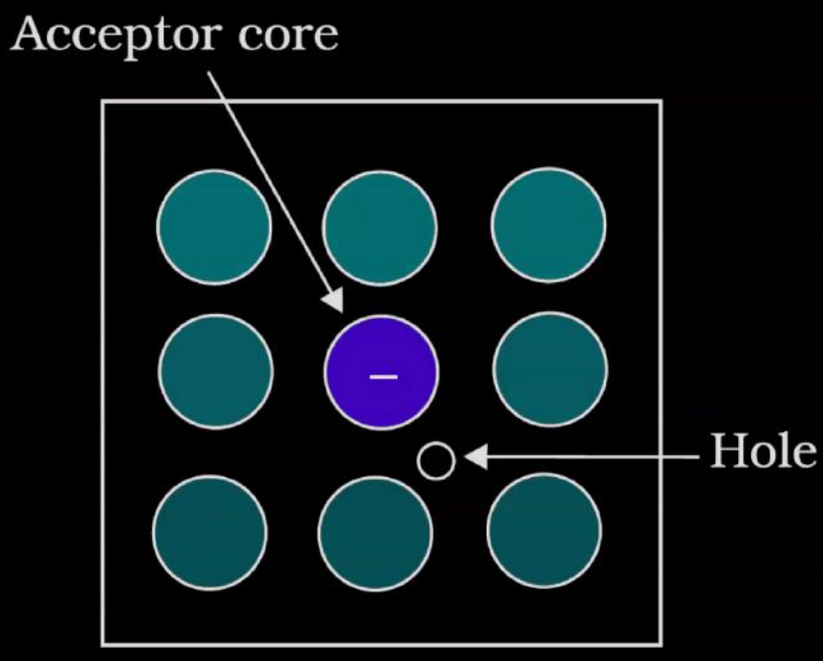
$n_h = (2.25 \times 10^{32})/(5 \times 10^{22})$

$\sim 4.5 \times 10^9$ m^{-3}

(1 ppm $= 10^{-6} \Rightarrow N_D = 5 \times 10^{28} \times 10^{-6} = 5 \times 10^{22}$)

P TYPE SEMICONDUCTORS:





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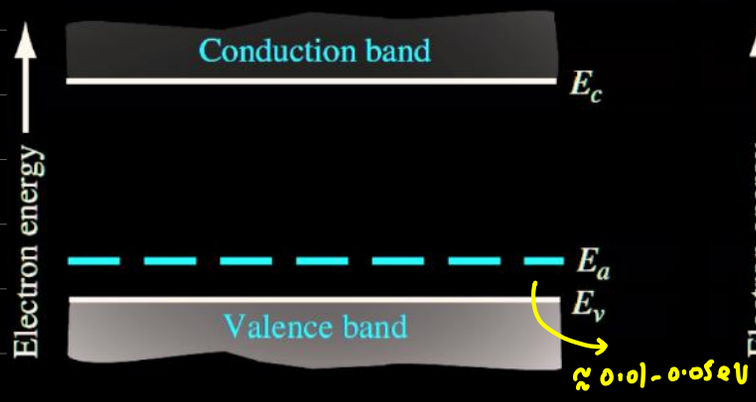

When a P type (trivalent) impurity is added to a semiconductor it is able to form a covalent bond with three silicon atoms but it has one lesser electron so it can accept one electron from a nearby silicon atom, thus creating a hole in the crystal. The fermi level of the acceptor is very close to the top of the valence band so electrons can very easily be accepted by the acceptor atoms. In this type of semiconductors, number of holes far exceeds the number of electrons and number of electrons is much smaller than that in an intrinsic crystal. Here also similar equations can be made for calculation of n_e and n_h as we did for N type impurities i.e:

$\boxed{n_e n_h = n_i^2}$ and $\boxed{n_h - n_e = n_A}$ A: Acceptor Here also with typical doping numbers, we can mostly use the equation $n_h \approx n_A$ and thus avoid solving the actual quadratic equation.

→ Since trivalent impurities accept an electron we have negatively charged ions in the crystal (like Al^- or In^- etc).

→ Overall the crystal is neutral since we added neutral atoms somewhere there is Al^+ and somewhere there is an extra e^- on Si^-.

→ In case of simultaneous doping with acceptor and donor impurities we can write $n_e - n_h = n_D - n_A$ and $n_e n_h = n_i^2$.

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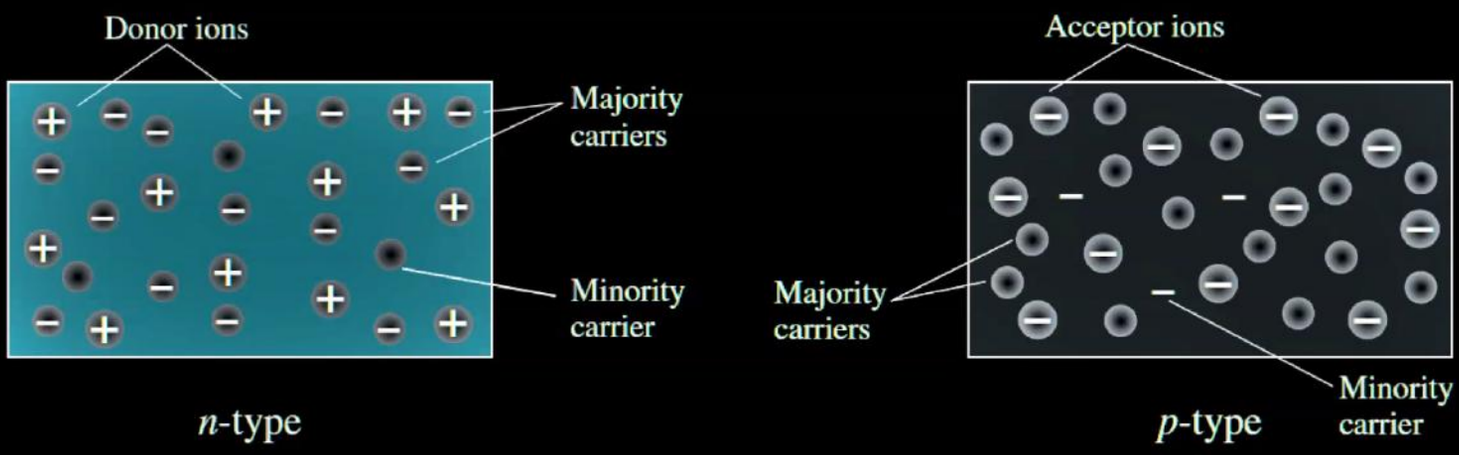
Majority And Minority Carriers:



In an n type crystal, electrons are known as majority carriers while holes are called minority carrier (because number of electrons is much more than that of holes) similarly in P type crystals holes are called majority carriers while electrons are called minority carriers. In a typical scenario, the current corresponding to majority carriers is in mA while that associated with minority carriers will be in micro amperes.

Mobility: Drift velocity produced per unit electric field is called mobility of the charge carriers. Typically holes have smaller mobility as compared to that of electrons. Mathematically, mobility, $\mu = v_d / E$ thus $\mu_e = v_e / E$ and $\mu_h = v_h / E$ where v_e and v_h are drift velocities of electrons and holes. The net current in the crystal is sum of the currents due to both types of charge carriers. Thus:

$$i = n_e e A v_e + n_h e A v_h = eA [n_e E \mu_e + n_h E \mu_h]$$

Thus

$$i = E e A [n_e \mu_e + n_h \mu_h], \text{ for intrinsic semiconductors, } n_e = n_h = n_i \text{ thus,}$$

$$\boxed{i = E e A n_i [\mu_e + \mu_h]} \leftarrow \text{Intrinsic.}$$

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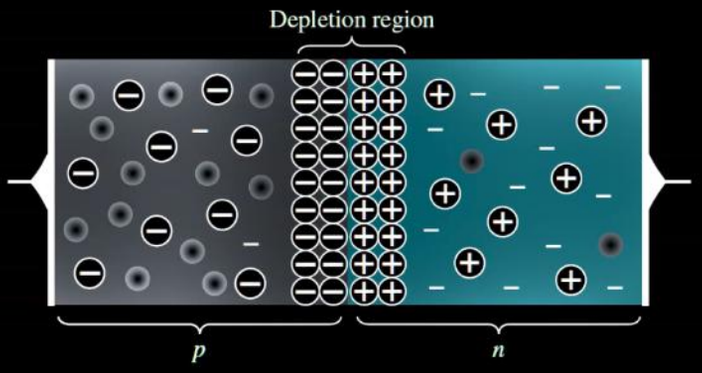
PN JUNCTION:

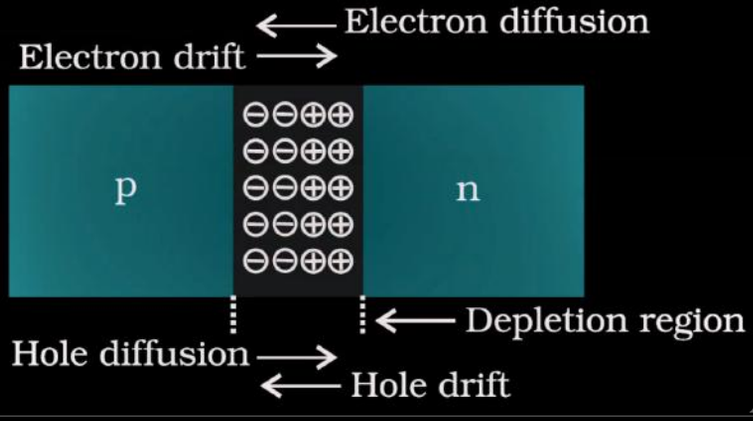


When part of a silicon crystal is doped with p type impurity and adjacent part is doped with n-type impurity we get p-n junction (see figure). Since n side has large electron concentration and p side has large hole concentration, due to difference in concentrations, holes diffuse from p side to n side and electrons diffuse from n side to p side, thus near the junction, electrons and holes combine with each other to produce a region which is practically devoid of charge carriers. This region is known as depletion region (or depletion layers). Due to depletion, we are left with $A\ell^-$ ions on p side and P^+ ions on the n side thus creating an electric field which tries to oppose the diffusion process.

DIFFUSION CURRENT: At the junction there are two competing processes happening dynamically. Due to concentration gradient, movement of charge carriers causes a current from p side to n side while due to electric field, there is a reverse motion of the charge carriers. The motion of charge carriers due to concentration gradient constitutes the diffusion current.

DRIFT CURRENT: As described above, because of diffusion, a net electric field is developed in the depletion region which pulls the holes towards p side and electrons towards n-side. This type of motion of charge carriers constitutes drift current.

The adjacent figure provides a graphical picture of the two competing processes. In equilibrium, two processes cancel the effect of each other and the net current is zero.



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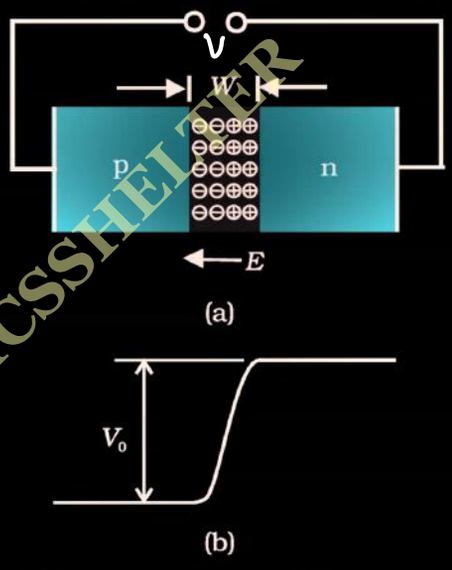
→ For functioning of a p-N junction as described above we can't simply join a p type and n type crystal slabs physically. For functioning like a p-n junction as described above, we need integrity of crystal at molecular bonds level, so same silicon crystal is partly doped with n type and p type impurities.

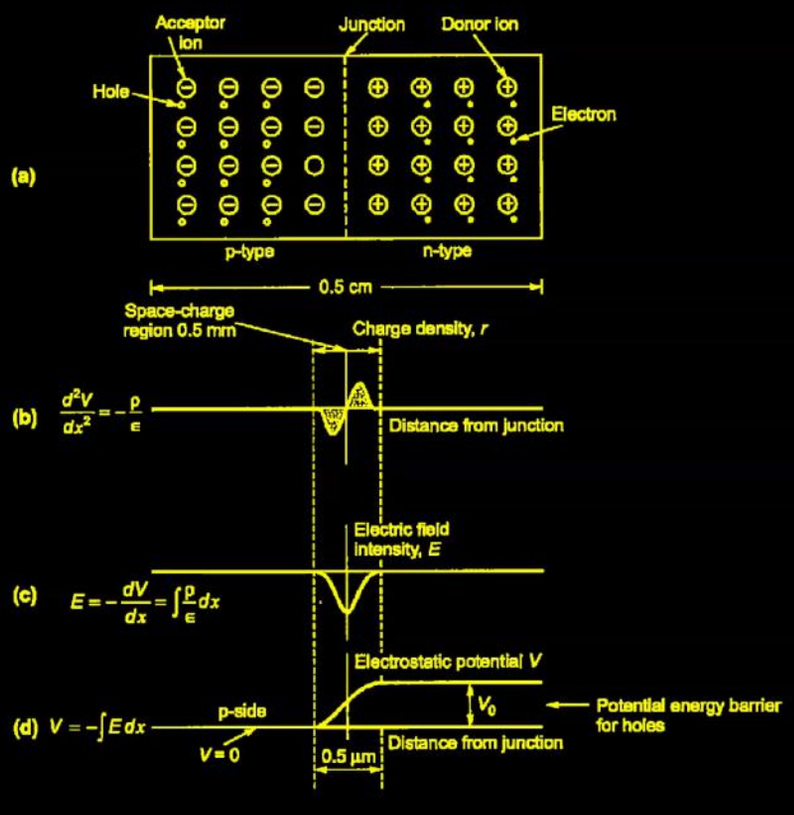
→ A p-n junction formed using same silicon crystal is known as a diode which is an important electronic circuit element.

Potential Barrier (No Bias):

When no battery is connected across a p-n junction, the diode is said to be unbiased. Due to presence of an electric field, the p side of the crystal is at a lower potential than the n side with a potential difference of about 0.7 Volts. This difference in potential is known as potential barrier and prevents spontaneous flow of current from p side to n side. (In fact depletion region has a very high resistance also due to absence of charge carriers. The adjacent figure shows variations of some other physical quantities in the depletion region namely, the charge density, electric field intensity and the variation of potential energy. Notice that maximum electric field occurs at the junction while the maximum charge density occurs a little far away from the junction. (Recall that $E = -dV/dx = \rho/\epsilon$).



(a) Diode under equilibrium (V = 0), (b) Barrier potential under no bias.



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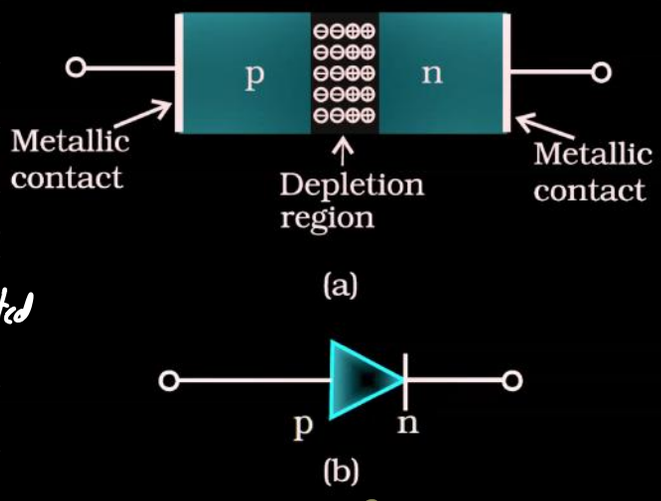
SEMICONDUCTOR DIODE:

A p-n junction provided with metallic contacts for connecting across a battery forms a semi-conductor diode.

The symbol for diode is also represented in the figure (b). The p side is represented by an arrow head pointing towards n side while n side is represented by a bar. Diode is a very important electronic circuit element and we will understand its behaviour in circuits in upcoming sections.

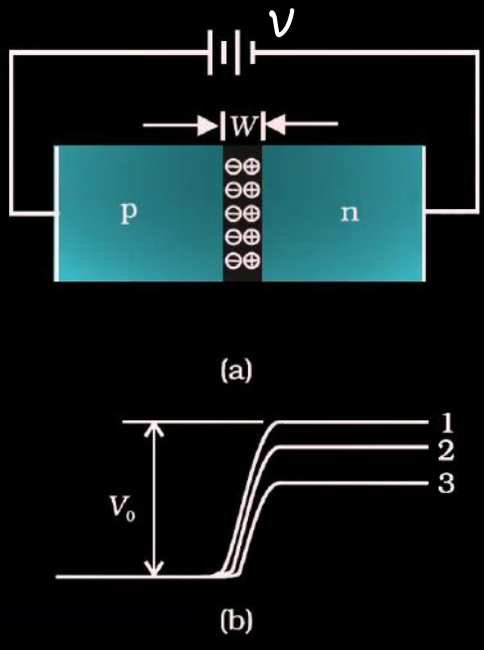


(a) Semiconductor diode,
(b) Symbol for p-n junction diode.

DIODE UNDER FORWARD BIAS (MINORITY CARRIER INJECTION):

When the positive terminal of the battery is connected to the p side and negative terminal the n side the diode is said to be forward biased. In semi-conductors it is often convenient to think of negative and positive terminals of a battery as electron and hole injectors respectively. Thus here electrons get injected from n side in to p side (where they are minority) and holes get injected from p side to n side. Forward biasing reduces the width of depletion region because battery supplies the charge carriers which are deficient on either side of depletion layer.

As the battery voltage is increased, the depletion layer width becomes smaller and smaller and the barrier height also keeps decreasing.



(a) p-n junction diode under forward bias,
(b) Barrier potential (1) without battery, (2) Low battery voltage, and (3) High voltage battery.

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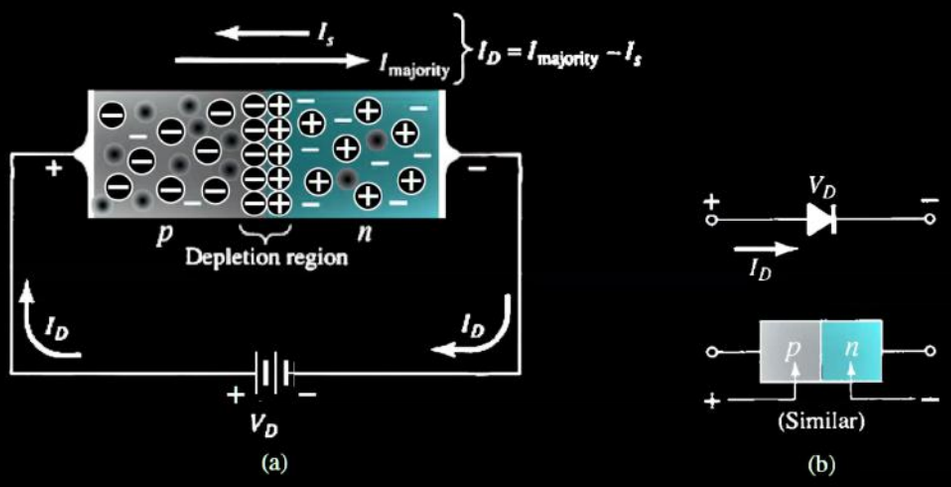
Here is a more detailed Graphic illustrating behavior of a diode in forward bias. (I_s is reverse saturation current i.e current due to diffusion of minority carriers).

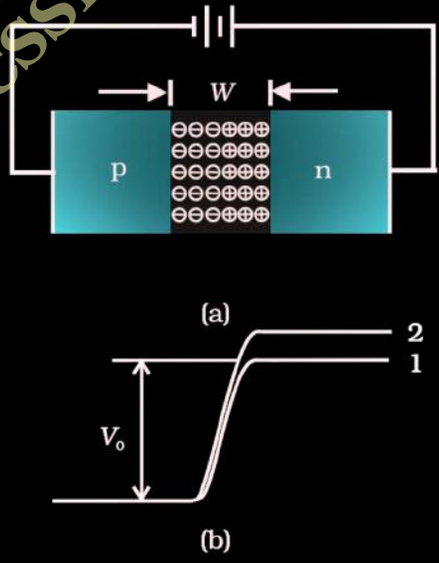


Forward-biased p–n junction: (a) internal distribution of charge under forward-bias conditions; (b) forward-bias polarity and direction of resulting current.

→ In forward bias the net current is along the diffusion current when the diode was unbiased. → Here effective barrier height is $V_0 - V$ (V: Applied battery, V_0: Unbiased).

Diode Under Reverse Bias:

When p side is connected to negative terminal of the battery and n side is connected to the positive terminal the diode is said to be reverse biased. Reverse bias is unfavorable to the flow of majority carriers (though favorable to the flow of minority carriers). Thus in reverse bias current is of the order of micro amperes. The depletion layer width increases here and so does the height of the potential barrier. Here the effective barrier height is $V_0 + V$ where V is the applied battery and V_0 is barrier height of unbiased diode. Here the electric field at the junction is such that if during random motion some majority carrier get in to depletion region it gets swept back to its majority region. The current in reverse bias does not change significantly with applied voltage because it is due to diffusion of minority carriers (even a small voltage is enough to sweep them across the junction) The current is not limited by voltage but it is limited by concentration of minority carriers on either side of the junction. The diode current



(a) Diode under reverse bias,
(b) Barrier potential under reverse bias.

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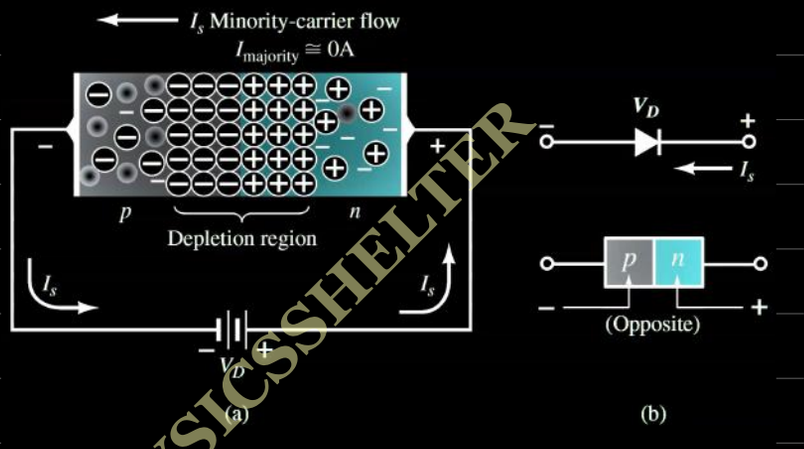
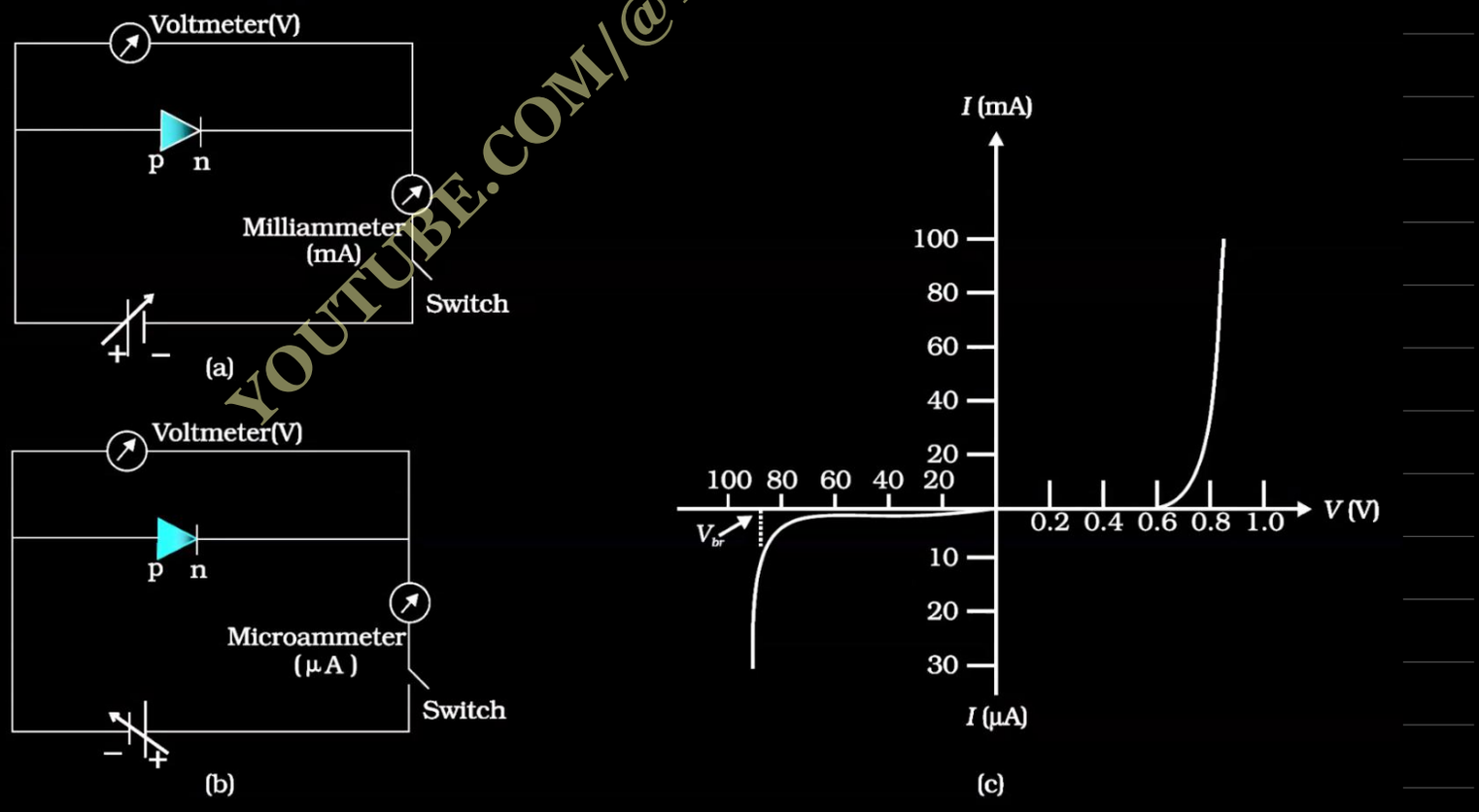
in reverse bias is essentially voltage independent up to a certain critical reverse bias voltage (V_{BR}) also known as breakdown voltage. We will learn about its mechanisms in an upcoming section. Beyond breakdown voltage even a small change in voltage will lead to large changes in current causing overheating and destroying the diode therefore sometimes external current limiting circuitry is used to protect the diode.

A more detailed graphic is presented in adjacent figure for easy understanding of current flow mechanism in a diode under reverse bias.



Reverse-biased p–n junction: (a) internal distribution of charge under reverse-bias conditions; (b) reverse-bias polarity and direction of reverse saturation current.

DIODE CHARACTERISTICS :



Experimental circuit arrangement for studying *V-I* characteristics of a p-n junction diode (a) in forward bias, (b) in reverse bias. (c) Typical *V-I* characteristics of a silicon diode.

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A graph between current and voltage for any electronic device is called its characteristic curve. The circuit set up for studying the characteristics in forward and reverse bias are shown in figures (a) and (b) while the characteristic curve is shown in figure (c). Note that current scales are not same on positive and negative side because in forward bias current is of the order of mA while in reverse bias it is in μA. The battery is connected to the diode through a potentiometer or a rheostat so that voltage may be varied. Also note that in forward bias we use a milliammeter while in reverse bias we use a microammeter.

FORWARD CHARACTERISTICS: As the biasing voltage is gradually increased from zero, the current increases negligibly till voltage across the diode crosses a certain value. After this voltage diode current increases significantly (exponential function), even for a very small increase in biasing voltage. This voltage is called the knee voltage or cut-in voltage (≈0.7 Volts for silicon and 0.3 Volts for germanium. (0.2 Volts for germanium according to NCERT)). For a typical diode, as per NCERT, current can be expressed as:

$$I = I_0 e^{\left(\frac{eV}{2kT} - 1\right)}$$

, I_0: Reverse saturation current, V: Voltage
T: Temperature, k: Boltzmann constant).

REVERSE CHARACTERISTICS: In reverse bias the current is very small and remains nearly constant with change in biasing voltage. This is called reverse saturation current. If the reverse bias is increased beyond a certain point the diode enters the breakdown region and current begins to increase suddenly. Most diodes are not used beyond the reverse saturation region except for some special purpose Zener and Avalanche diodes discussed in a later section.

DYNAMIC RESISTANCE: For typical applications we operate the diode in a certain small region of the characteristic graph. The ratio of change in voltage to change in current near this point

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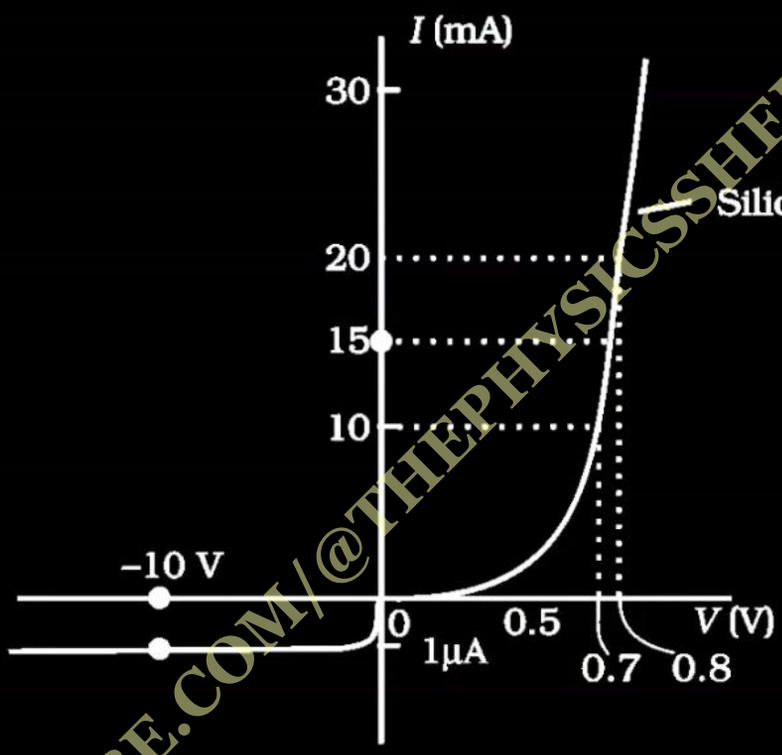
is called dynamic resistance. Mathematically $r_d = \Delta V/\Delta I$

In reverse bias since current remains constant, the resistance is simply taken as V/I.

EXAMPLE: NCERT:

Example 14.4 The *V-I* characteristic of a silicon diode is shown in the Fig. 14.17. Calculate the resistance of the diode at (a) I_D = 15 mA and (b) V_D = –10 V.



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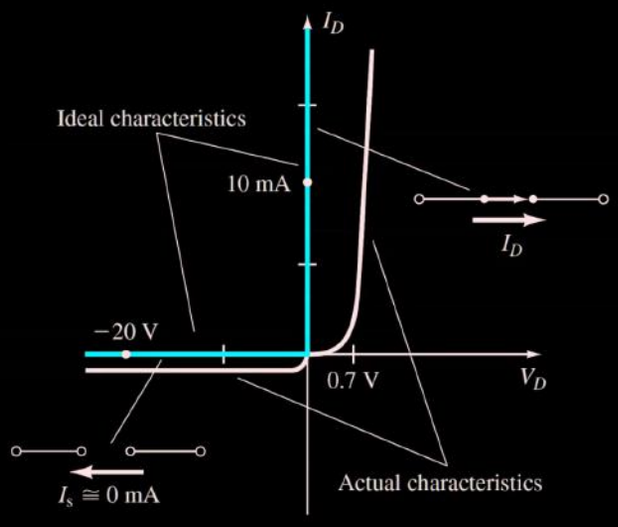
Solution Considering the diode characteristics as a straight line between I = 10 mA to I = 20 mA passing through the origin, we can calculate the resistance using Ohm's law.

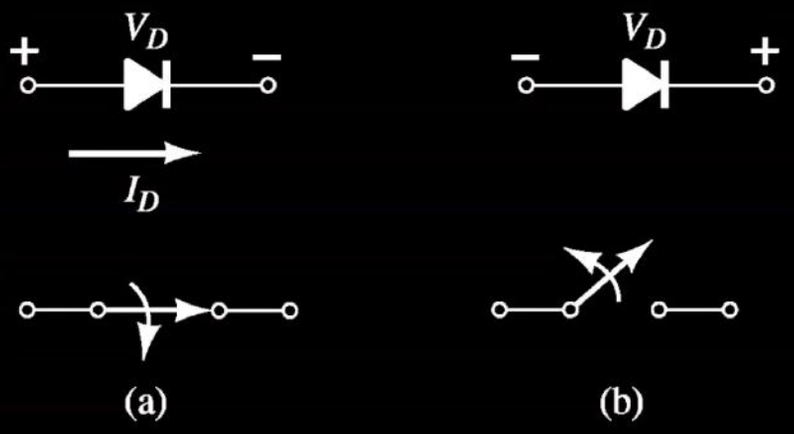
(a) From the curve, at I = 20 mA, V = 0.8 V; I = 10 mA, V = 0.7 V
$r_{fb} = \Delta V/\Delta I$ = 0.1V/10 mA = 10 Ω

(b) From the curve at V = –10 V, I = –1 μA,
Therefore,
r_{rb} = 10 V/1μA= 1.0×10^7 Ω

Ideal Diode :



Ideal versus actual semiconductor characteristics.



Ideal semiconductor diode: (a) forward-biased; (b) reverse-biased.

In an ideal diode we assume that in forward bias, the diode acts like a short circuit (offers zero resistance) while in reverse bias we assume it to be like an open switch (infinite resistance).

Handling Ideal Diode As A Circuit Element :

An ideal diode in a circuit could either be forward biased or reverse biased and we may not know it beforehand. So to find out we can use one of the following procedures:

Method I:

① Assume diode to be short circuited
② Find current through the diode under this assumption
③ If current is in forward direction assumption is correct otherwise repeat again using open circuit assumption.

Method II:

① Assume open circuit
② Find potential difference across the diode under this assumption
③ If diode is reverse biased our assumption is correct otherwise we have to repeat assuming short circuit.

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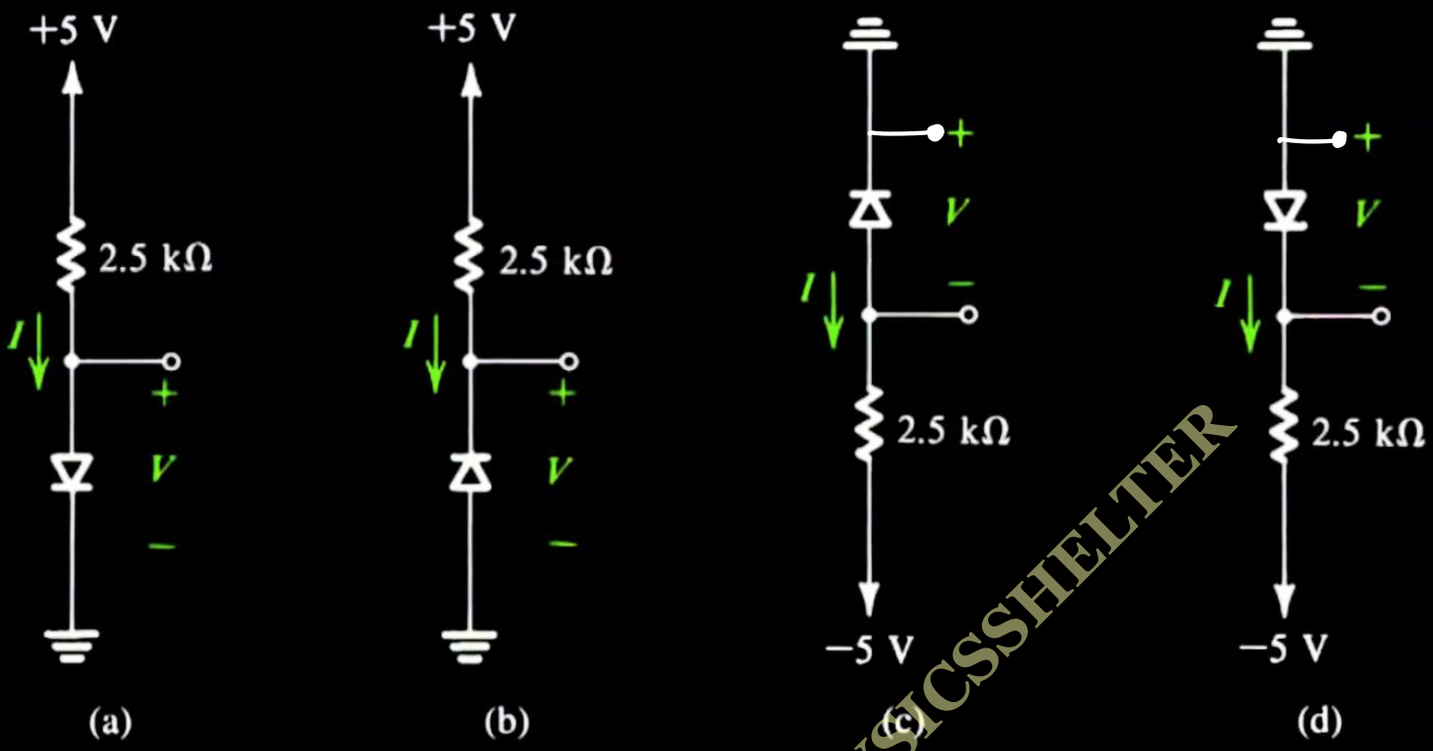
Example: Find I and V in following circuits:



Solution: (a) Assuming forward bias, $I = 5/2.5\times10^3 = 2\,mA$, $V = 0$

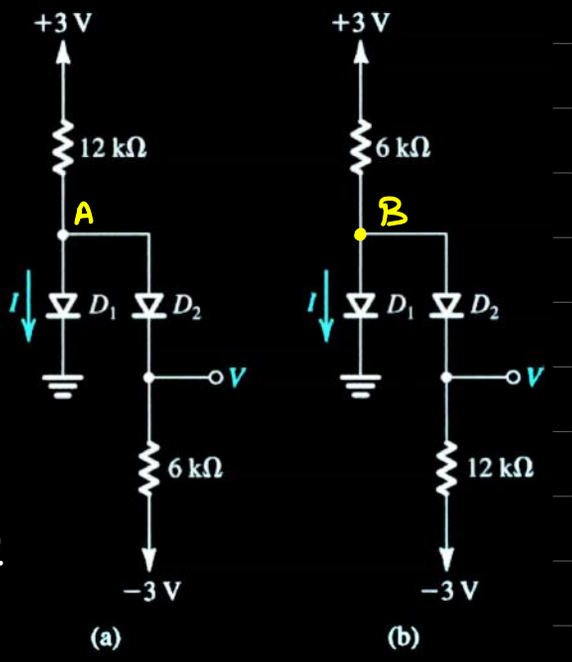
(b) Assuming reverse bias, $I = 0$, $V = 5$ Volts

(c) Assuming reverse bias, $I = 0$, $V = +5$ Volts

(d) Assuming forward bias, $I = 2\,mA$, $V = 0$ Volts.

Example:

For the diode circuit in adjacent figure, find the values of currents in the indicated branches.



Solution: (a) Assuming both diodes short,

$V_A = 0 \Rightarrow I_{12} = 3/12 = 1/4\ mA$,

$I_6 = (0-(-3))/6 = 1/2\ mA$

$\Rightarrow I = 1/4 - 1/2 = -1/4\ mA \Rightarrow$ Assumption failed.

Now assuming D_1 open & D_2 short,

$I_{12} = I_6 = \frac{6}{12+6} = 1/3\ mA$, $V_A = 3 - 12\times\frac{1}{3}$

$= -1$ Volt. $\Rightarrow D_1$ reverse confirmed.

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Thus $I = 0$ Amps.

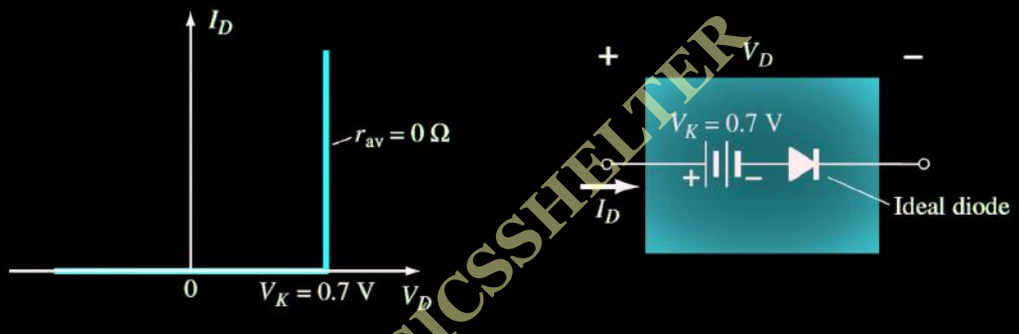
(b) Assuming both forward, $V_D = 0 \Rightarrow I_6 = 1/2$ mA, $I_{12} = 1/4$ mA

$\Rightarrow I = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ mA $\Rightarrow$ Assumption holds.

Thus $I = 1/4$ mA Amps., $V = 0$ (shorted with ground).

Silicon Diode Simplified Equivalent:

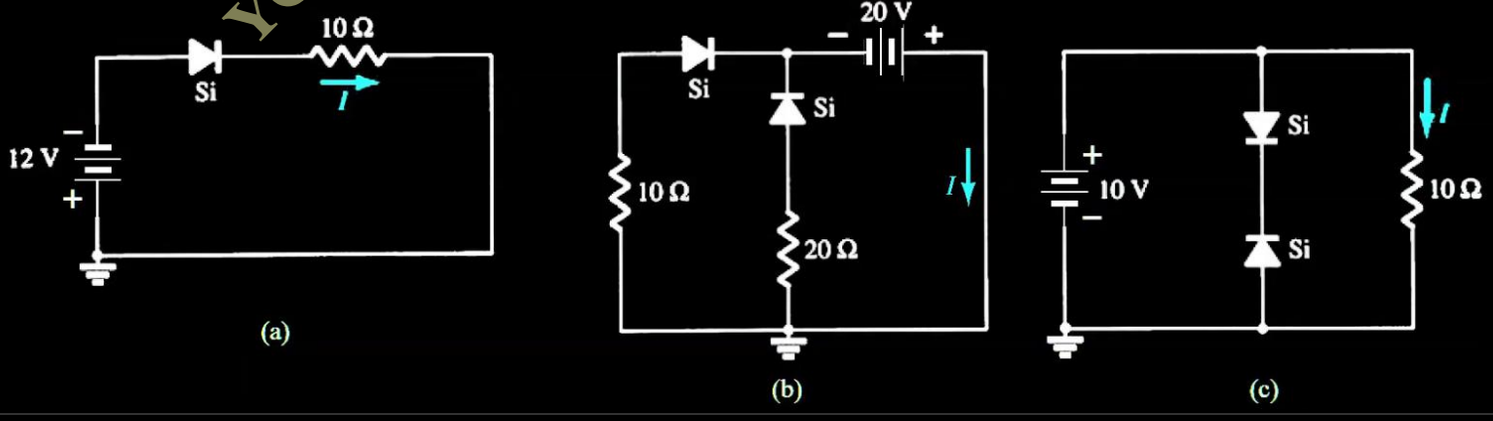
For silicon diodes, we can assume that until forward bias voltage is less than 0.7 Volts, there is no current. Thus we can treat it like a combination of reverse 0.7 Volt battery and an ideal diode as seen in the figure above. Here we will treat 0.7 Volts just like an external battery and rest of the procedure is just like that in ideal diodes. (For Germanium diodes the reverse battery is taken as 0.3 Volts).



Simplified equivalent circuit for the silicon semiconductor diode.

Example:

Find current I through each of the shown diode circuits:



Solution: (a) Diode is reverse biase (0V = −12.7 Volt) $\Rightarrow I = 0$

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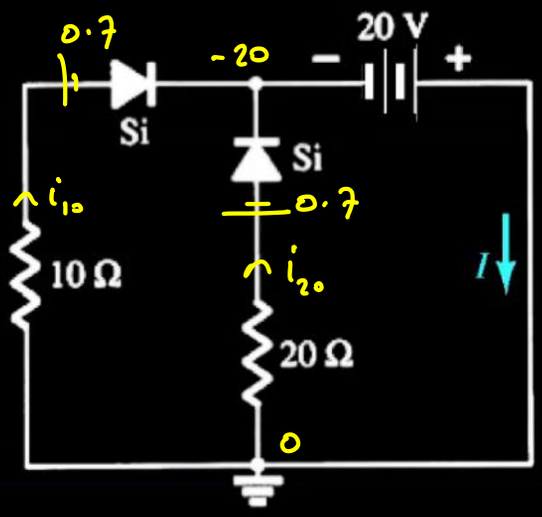
(b) Assuming both diodes forward,

$i_{20} = 19.3/20 = 0.965\ A$

$i_{10} = 19.3/10 = 1.93\ A$

$\Rightarrow I = i_{20} + i_{10} = 2.895\ A$ Ans.



(c) Clearly both diodes are head on against each other so no current can flow through the middle branch.

Thus, $I = 10/10 = 1\ A$ Ans.

EXAMPLE:

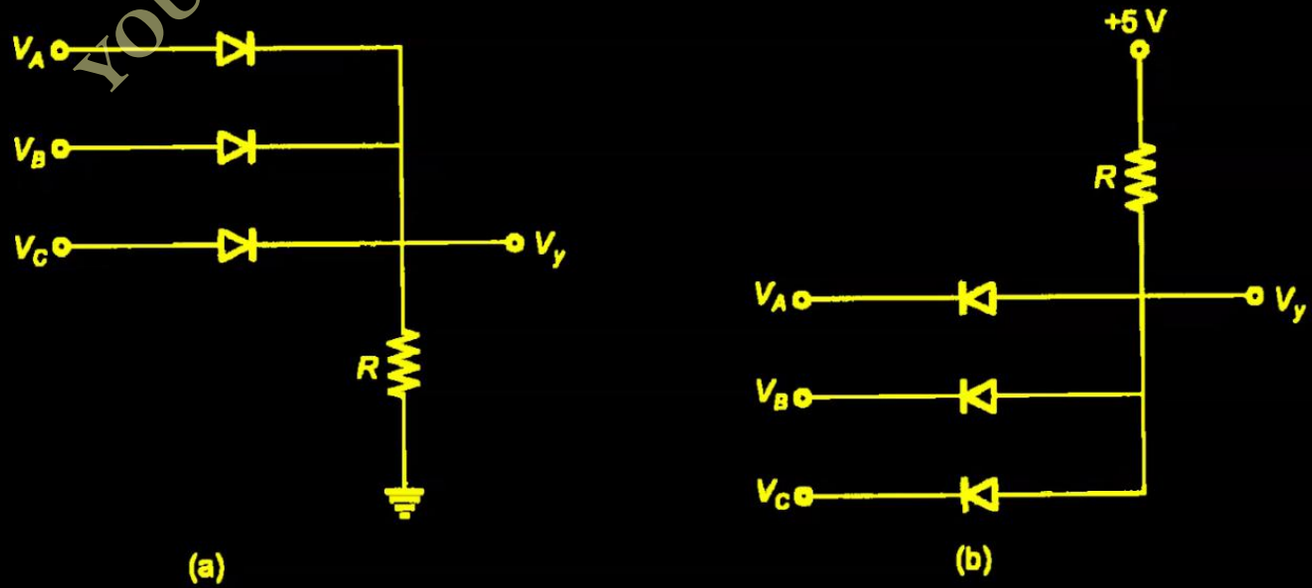
Find V_o for the shown circuit. Assume 0.2 V for germanium.

Solution: Assuming both diodes forward, $I_{10} = \dfrac{(12 - 0.7 - 0.2)}{(10 + 2)}$

$\Rightarrow I = 0.925\ mA \Rightarrow V_o = I \times 10$

$\Rightarrow V_o = 9.25$ Volts Ans.

DIODE LOGIC GATES:



Diode logic gates (a) OR gate (b) AND gate

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In binary logic circuits voltages are just assigned binary values say High (5 Volts say) and low (say 0 volt). Now in circuit A recognize that if any one of the voltages V_A, V_B or V_C is high, The V_y will become high because the diode with high voltage will act as a short circuit. This type of situation where a high voltage in any one of the several places (input) guarantees a high voltage in another place (output) is called an OR gate.

In circuit (b), if any of V_A, V_B or V_C is low then that diode will act as a short circuit making V_y also low. To have a high value of V_y, each of V_A, V_B and V_C must be high to ensure an open circuit of all the three diodes. If all the three diodes are reverse biased, no current will flow through the resistor and accordingly V_y will be 5V i.e high. This type of situation when each of the several voltages (inputs) must be high to ensure a high value at another place (output) is called an AND gate.

→ Note that in circuit (a) if each of the V_A, V_B & V_C were different then only one of the diodes will be forward bias (One with most positive value of voltage). (The junction can't be at three different potentials simultaneously).

→ Similarly in circuit (b) if V_A, V_B and V_C were all negative, only the diode with lowest voltage (most negative) will be forward biased. (Reason similar to that in figure (a)).

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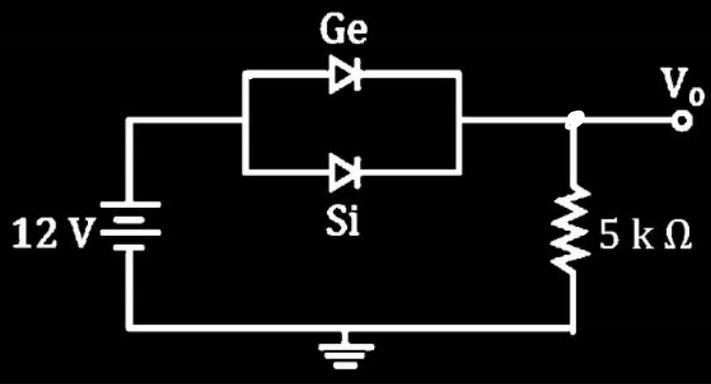
EXAMPLE (JEE MAIN):

For the shown circuit find the change in value of V_0 if the Germanium diode were reversed.



Solution: In forward bias, diodes maintain a constant potential difference in the simplified model. So both diodes can't be simultaneously forward biased (Reverse voltage is different for both).

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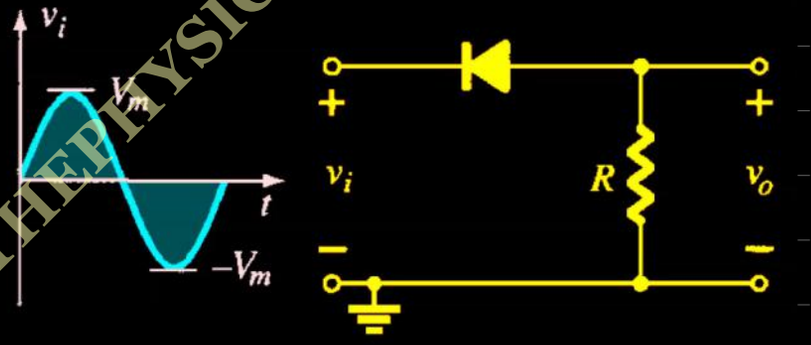
Thus initially only Ge diode will be forward biased and accordingly $V_o = 12 - 0.3 = 11.7$ Volt.
When Ge diode is reversed, silicon diode becomes forward biased and new voltage $V_o' = 12 - 0.7$ i.e 11.3 Volt thus V_o reduces by 0.4 volts thus.

Sketching Ac Response To Diode Circuits:

In this category of problems as the input voltage changes, we have to figure out the range of input voltage for which the diode will remain forward biased and reverse biased and then use logic to sketch the output.

Example:

Sketch the output for the given circuit for the given input voltage.



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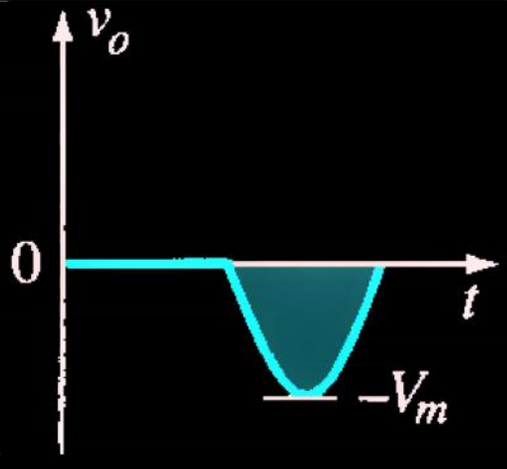
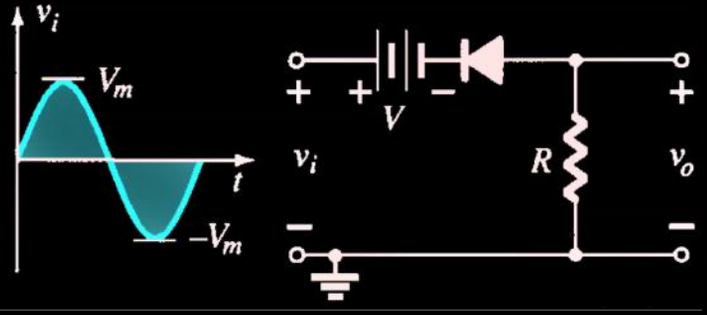
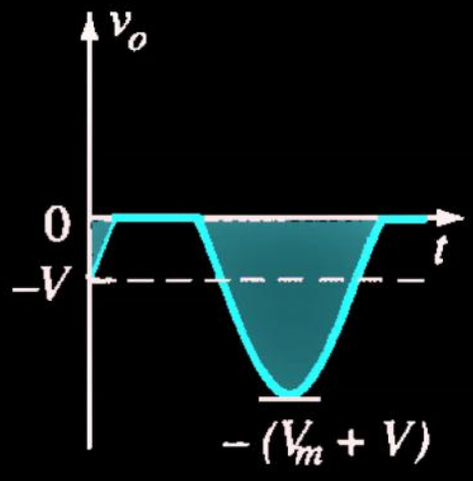
Solution: Here for the positive half cycle, diode remains open circuit so current through R becomes zero and so also V_o. For negative half cycle, diode is short and V_o becomes equal to V_i (see figure).



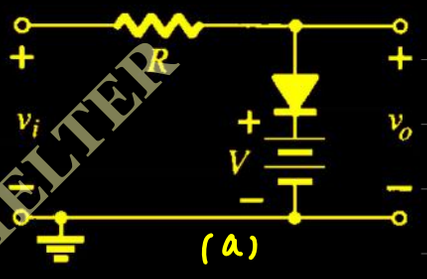
Example: Sketch the output signal for the given circuit with sinusoidal input signal.

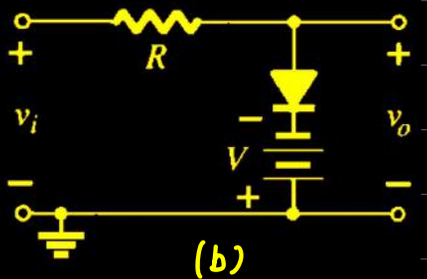


Solution: Here until input voltage is less than $+V$, diode remains short and output voltage is $V_i - V$, when input voltage is more than $+V$, diode is open circuit and current through R becomes zero and so also the output voltage.

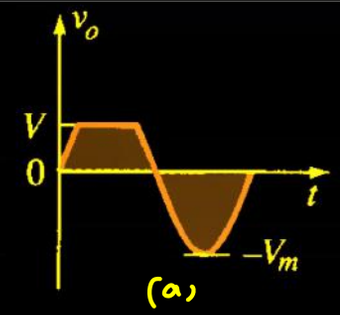


EXAMPLE: Sketch the output voltage for the shown circuits for the standard sinusoidal input signal (same as previous example).

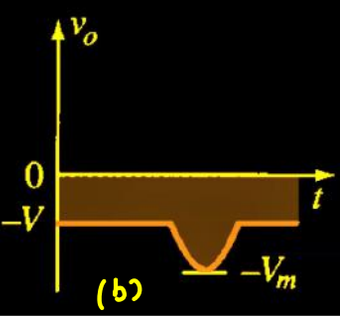




Solution: (a) In this case until input voltage is more than $+V$, diode remains short and output voltage is simply $+V$. When input becomes less than $+V$, diode becomes open and actual voltage is then same as input voltage



(b) Here until input is more than $-V$, the diode remains short and output voltage is simply $-V$ when input voltage becomes less than $-V$, diode becomes open and output voltage becomes same as input voltage.



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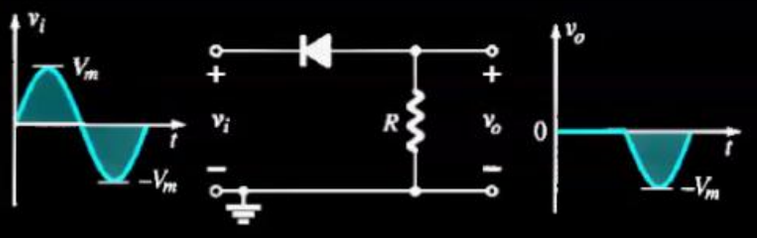
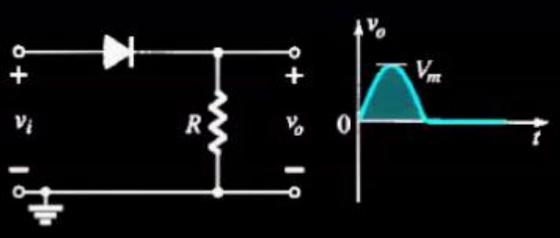
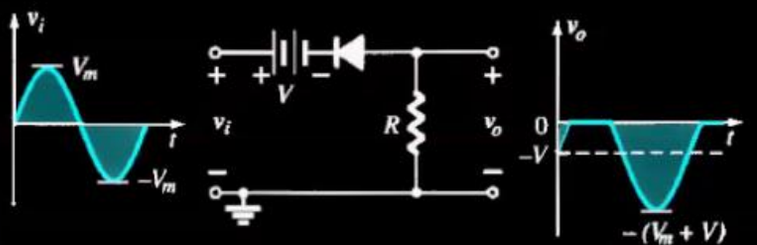
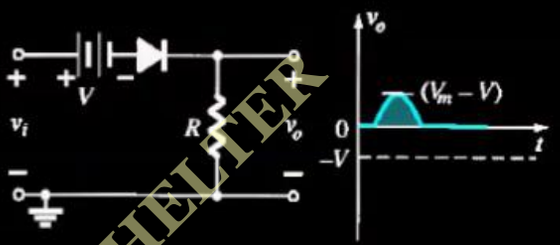
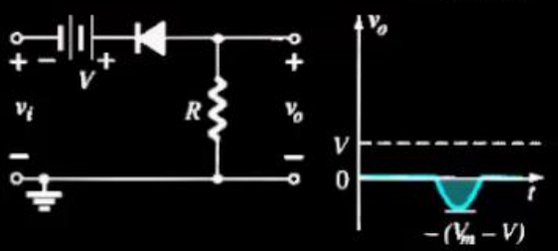
Think! For each of the shown circuits, verify the shown output (standard input).

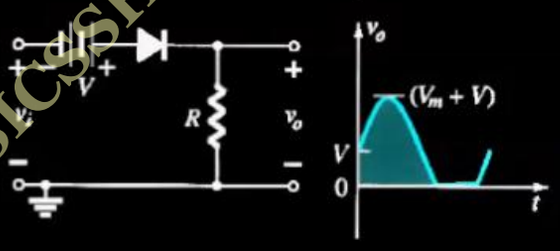


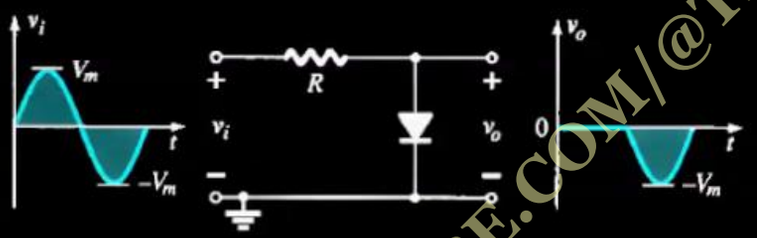


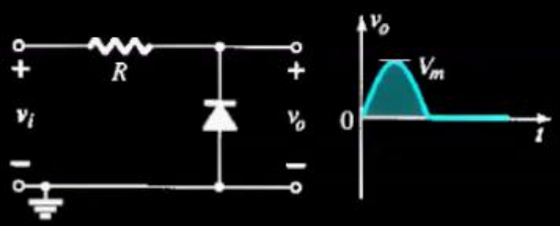


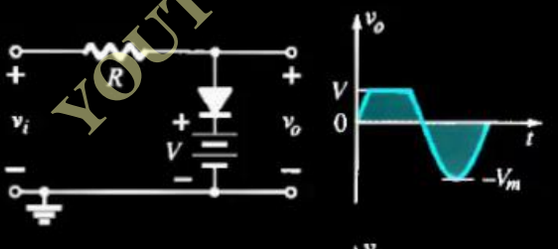


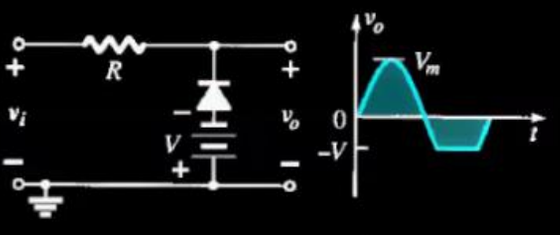


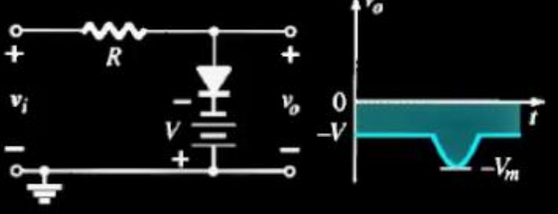


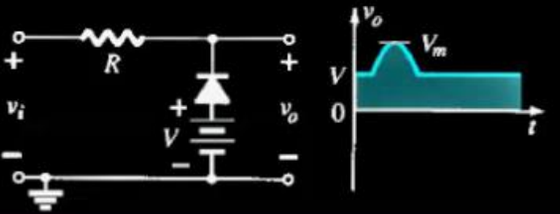


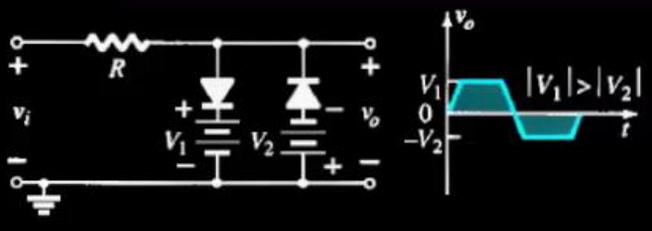












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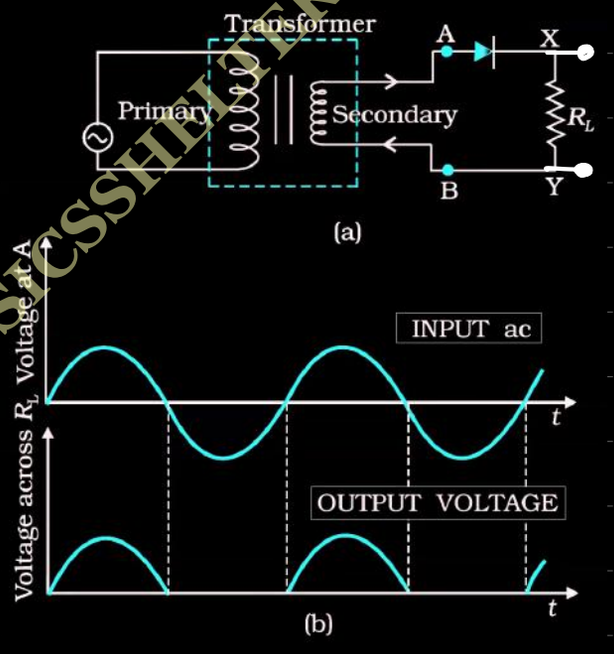
Using Junction Diode As A Rectifier:

The process of converting an A.C. signal into a dc signal is called rectification. Since diodes exhibit unidirectional current flow practically, they can be conveniently used to rectify the A.C. signals. Here we will study various diode configuration for rectification namely half wave rectifier, full wave rectifier (centre tap) and Bridge rectifier (full wave).

Half Wave Rectifier:

A half wave rectifier is so named because its output contains only half of the sinusoidal wave and cuts off the other half. (See adjacent figure (b)). The schematic circuit for half wave rectification is shown in figure (a). Here due to EMI, the terminal A of the secondary alternately becomes positive and negative in response to sinusoidal excitation by the primary. The diode conducts only when A is at a positive potential relative to B and it is open for negative values. Thus current flows through R_L only in downward direction during positive cycles. The output is measured across load resistance R_L. Since no current flows through R_L during the negative cycle, there is no negative cycle in the output.



(a) Half-wave rectifier circuit,
(b) Input ac voltage and output voltage waveforms from the rectifier circuit.

DC Equivalent Current:

The average current flowing per unit time is called DC equivalent current. For a piecewise periodic function if T_1 and T_2 are time periods of two pieces and I_1 and I_2 are rectified average currents for these two periods then net rectified average current is given as $I_{dc} = \frac{I_1 T_1 + I_2 T_2}{T_1 + T_2}$. For our output, the output has two equal time parts in each cycle with respective rectified average values as $I_m/2\pi$ and 0 (See the output

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wave).

Thus $I_{dc} = \dfrac{(I_m \times \frac{2}{\pi}) \times T/2 + 0 \times T/2}{T} = I_m/\pi.$

Thus $\boxed{I_{dc} = \dfrac{I_m}{\pi} = \dfrac{V_m}{\pi R_L}}$ HWR $I_m \rightarrow$ peak current $V_m \rightarrow$ peak voltage.

RMS CURRENT: If I_{rms_1} and I_{rms_2} are rms currents for parts of a cycle having durations T_1 and T_2 then the effective rms current is given as $I_{rms} = \left[\dfrac{I_{rms_1}^2 \times T_1 + I_{rms_2}^2 \times T_2}{T_1 + T_2}\right]^{1/2}$

For our half rectified wave, $I_{rms_1} = I_m/\sqrt{2}$ $I_{rms_2} = 0$ thus,

$I_{rms} = \left[\left(\left(\dfrac{I_m}{\sqrt{2}}\right)^2 \times \dfrac{T}{2} + 0 \times \dfrac{T}{2}\right)/T\right]^{1/2} = \dfrac{I_m}{2} = \dfrac{V_m}{2R_L}$

Thus: $\boxed{I_{rms} = \dfrac{I_m}{2} = \dfrac{V_m}{2R_L}}$ HWR

RMS AC CURRENT: If i is the instantaneous current and I_{dc} is the dc current then rms value of $i - I_{dc}$ is called RMS A.C. current. It can be shown that:

$$\boxed{I_{AC} = \sqrt{I_{rms}^2 - I_{DC}^2}}$$

RIPPLE FACTOR: The ratio of I_{AC} to I_{DC} is called ripple factor.

Thus ripple factor:

$$\gamma = \frac{I_{AC}}{I_{DC}} = \sqrt{\left(\frac{I_{rms}}{I_{DC}}\right)^2 - 1}$$

Thus for half wave rectifier, ripple factor: $\gamma = \sqrt{\left(\dfrac{I_m/2}{I_m/\pi}\right)^2 - 1} = 1.21$

Thus ripple factor for Half rectified wave: $\boxed{\gamma = 1.21 = 121\ \%}$ HWR

→ Ripple factor is a measure of efficacy of a rectifier or a filter in producing a smooth output. Higher the ripple factor, more is the fluctuating AC component relative to the dc portion. For ideal dc waveform, $I_{DC} = I_{AC}$ and $\gamma = 0$.

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INPUT POWER: If the diode in the rectifier circuit has a forward resistance R_f then some power is lost as heat in the diode while another part is dissipated in R_L. The sum of powers dissipated in the diode and R_L is called input power (This is also the power supplied to the secondary coil). Thus input power $P_i = I_{rms}^2 (R_f + R_L)$.

DC OUTPUT POWER: The power output in R_L if the total charge delivered by the rectifier were supplied at a uniform rate (i.e pure dc current) is called dc output power. Thus $P_{dc} = I_{dc}^2 \times R_L$.

RECTIFIER EFFICIENCY: The ratio of dc output power to the input power is called rectifier efficiency. Thus, $\eta = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_L + R_f)}$

$$= \frac{I_{DC}^2}{I_{rms}^2}\left[\frac{1}{1 + R_f/R_L}\right] = \left(\frac{2}{\pi}\right)^2 \left[\frac{1}{1 + R_f/R_L}\right] \simeq \frac{40.6}{1 + R_f/R_L}$$

Thus $\boxed{\eta_{HWR} = \frac{40.6}{1 + R_f/R_L}}$ HWR

If dide is ideal, $R_f = 0$, $\eta_{HWR} = 40.6\ \%$.

RIPPLE FREQUENCY: The frequency of bumps in the rectified output voltage is called Ripple frequency. For a half wave rectifier it is same as the frequency of the input signal. Thus if AC input frequency is 50 Hz, Ripple frequency is also 50 Hz.

PEAK INVERSE VOLTAGE: During the negative cycle, the diode is reverse biased. The maximum value of this voltage is called peak inverse voltage. This must be less than the reverse breakdown voltage of the diode. To avoid the breakdown, sometimes, multiple diodes are used in series to divide the reverse voltage across the diodes and thus avoiding the breakdown.

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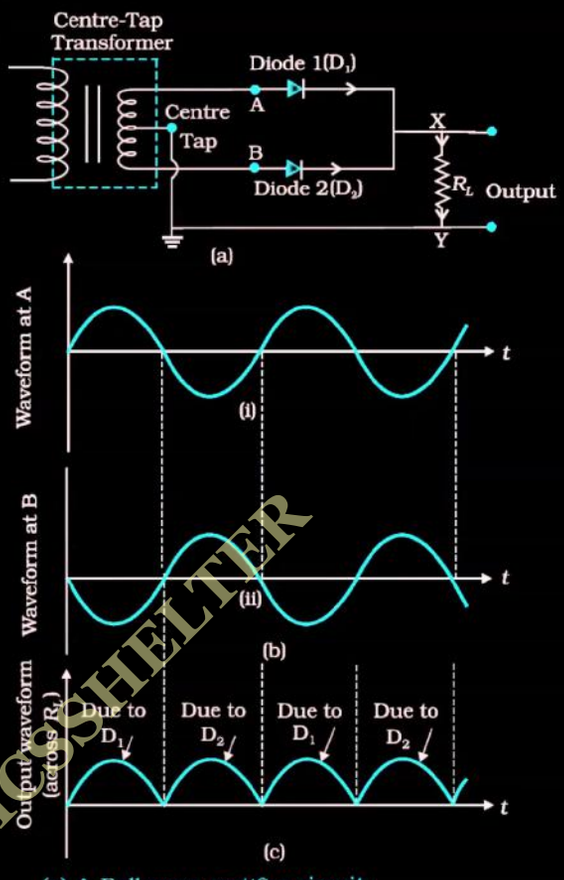
Full Wave Rectifier: Centre Tap:

Consider the circuit shown if figure (a). Here as the primary voltage oscillates, alternately points A and B will become positive thus alternately forwarding the upper and the lower diodes. Notice that In both cases, the current through R_L is always downwards. Thus output voltage across R_L is always positive irrespective of the positive or negative cycle on the input side. This type of rectifier is called a full wave rectifier.



(a) A Full-wave rectifier circuit;
(b) Input wave forms given to the diode D_1 at A and to the diode D_2 at B;(c) Output waveform across the load R_L connected in the full-wave rectifier circuit.

DC Equivalent Current: Here using standard theory of AC we can directly see that rectified average value of current will be

$$I_{DC} = \frac{2I_m}{\pi} = \frac{2V_m}{\pi R_L}$$ FWR

V_m: Peak output Voltage.

RMS Current: Once again using standard AC theory we know that here RMS current will be simply:

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{V_m}{\sqrt{2}R_L}$$ FWR

AC RMS Current: If i is instantaneous current and I_{DC} is the DC current, then rms value of $i - I_{DC}$ is called AC rms current. We can show this to be equal to: $I_{AC} = \sqrt{I_{rms}^2 - I_{DC}^2}$

The proof is as follows: $i = i_{AC} + i_{DC} \Rightarrow i_{AC} = i - i_{DC}$

$$\Rightarrow i_{AC}^2 = i^2 + i_{DC}^2 - 2ii_{DC}$$

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$$\Rightarrow I_{AC}^2 = i_{AC_{rms}}^2 = I_{rms}^2 + I_{DC}^2 - \left[2 I_{DC} \int_0^{2\pi/\omega} i\,dt\right] \cdot \frac{1}{2\pi}$$

$$= I_{rms}^2 + I_{DC}^2 - 2I_{DC}^2 = I_{rms}^2 - I_{DC}^2$$

Thus,

$$I_{AC} = \sqrt{I_{rms}^2 - I_{DC}^2} \qquad \text{QED (Same proof in HWR also).}$$

RIPPLE FACTOR: The ratio of dc current to the AC current is called ripple factor. Thus:

$$\gamma = \frac{I_{AC}}{I_{DC}} = \sqrt{\frac{I_{rms}^2}{I_{DC}^2} - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1}$$

$$\Rightarrow \boxed{\gamma = 0.48 = 48\%}_{\text{FWR}}$$

RECTIFIER EFFICIENCY (FWR): The ratio of output DC power to the input power is called Rectifier efficiency. Now:

Input Power, $P_i = I_{rms}^2 (R_f + R_L)$, R_f: Forward resistance of diode.

DC output Power, $P_{DC} = I_{dc}^2 (R_L)$. Thus rectifier efficiency:

$$\eta = \frac{P_{DC}}{P_i} = \left(\frac{I_{dc}}{I_{rms}}\right)^2 \frac{1}{1 + R_f/R_L} = \left(\frac{2\sqrt{2}}{\pi}\right)^2 \frac{1}{1 + R_f/R_L} \approx \frac{0.812}{1 + R_f/R_L}$$

Thus:

$$\boxed{\eta_{FWR} = \frac{81.2}{1 + R_f/R_L}\ \%}_{\text{FWR.}}$$

RIPPLE FREQUENCY: Here there will be two bumps in the rectified output in every input cycle. Thus if input frequency is 50 Hz, output ripple frequency will be 100 Hz. Thus $\boxed{\nu_R = 2\nu_i}_{\text{FWR}}$

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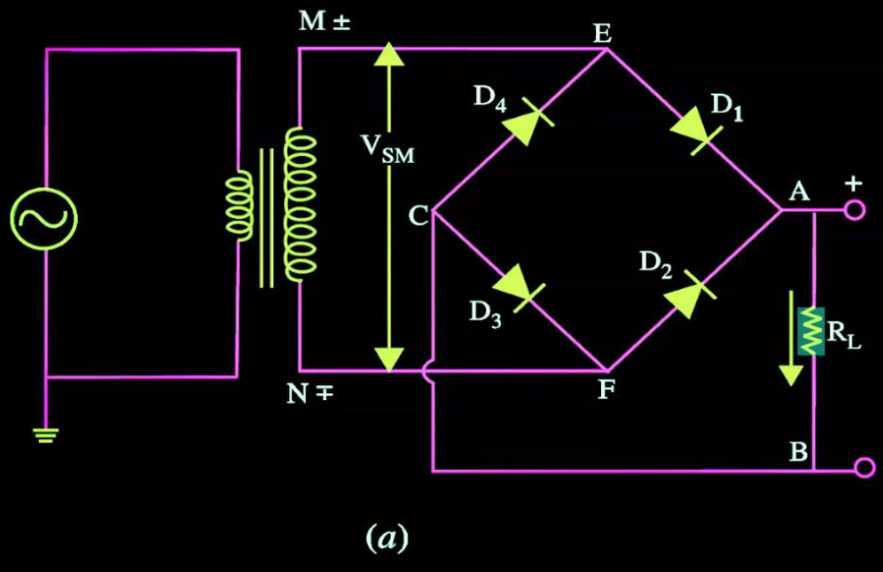
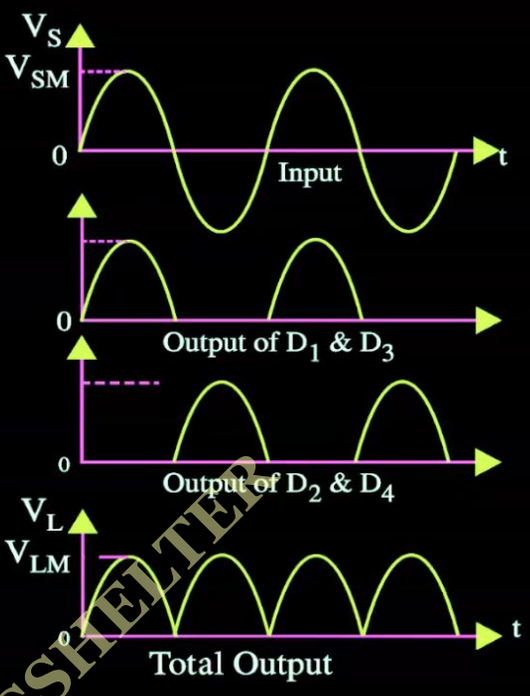
BRIDGE RECTIFIER:

V_{sm} : Max Secondary Voltage

V_{Lm} : Max Voltage Across R_L



(a)



(b)

The figure (a) above shows the schematic circuit of a Bridge rectifier. This is also a full wave rectifier. During the positive cycle, diodes D_1 and D_3 are forward biased while during negative cycle, D_2 and D_4 are forward biased. Thus current path during positive cycle is : M E D_1 A B C D_3 F N M and current path during negative cycle is N F D_2 A B C D_4 E M N, in both the paths, the current through AB is always in downward direction. Thus output voltage across AB is always positive.

→ Here also all the physical quantities defined for centre tap full wave rectifier are described by the same formulae.

→ Here peak inverse voltage across each diode will be half of V_{sm} because two diodes will act in series thus sharing half of the total potential drop.

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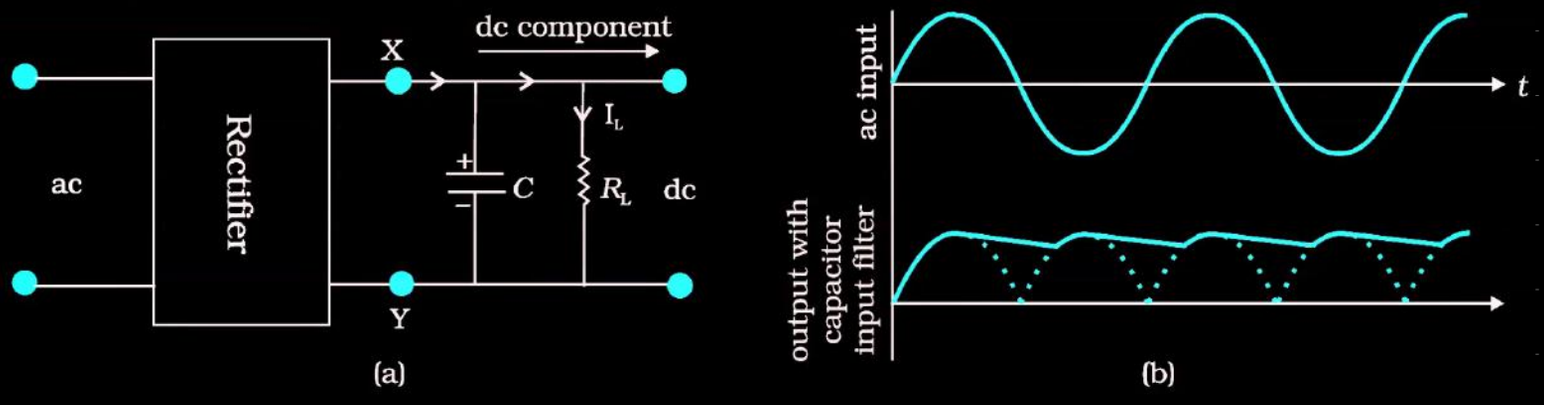
Rectifier With Capacitor Filter:



(a) A full-wave rectifier with capacitor filter, (b) Input and output voltage of rectifier in (a).

The rectified voltage is in the form of pulses of the half sinusoidal shapes. Though the output is unidirectional, it does not have a steady value like a nice dc from a battery. To get a steady dc from the pulsating voltage, normally a capacitor is connected in parallel with R_L (or sometimes an inductor in series with R_L). Since these additional circuits filter out the AC ripples, these circuits are called filter circuits.

Capacitor Filter Mechanism: The discharging time constant of the capacitor circuit ($R_L \times C$) is sufficiently large. When voltage output of rectifier is increasing, capacitor keeps getting charged and when the voltage begins to decrease (smaller than capacitor voltage) the diode becomes reverse biased and capacitor begins to discharge through R_L. Even though this discharging leads to an exponential decaying voltage, since the time constant of discharging is very high, we have a nearly flat graph. Once again when voltage starts rising, the capacitor again starts getting charged when diode becomes forward biased and whole cycle keeps repeating over and over. In this manner we are able to generate a less bumpy output voltage, closer to the one generated by a dc battery.

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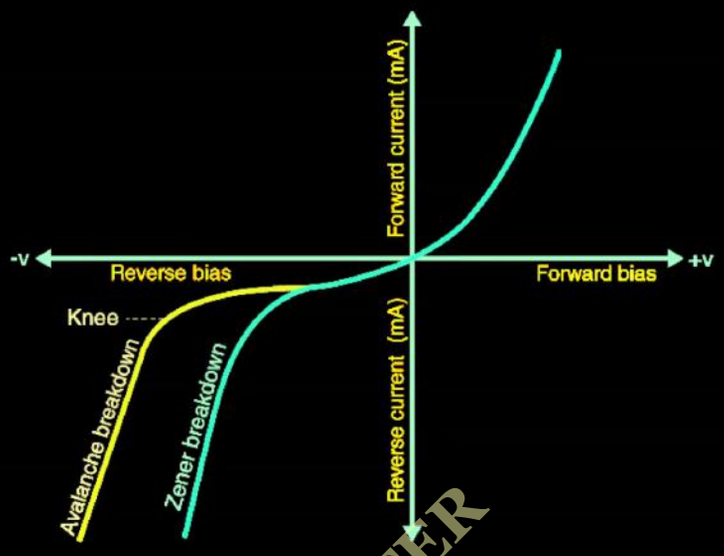
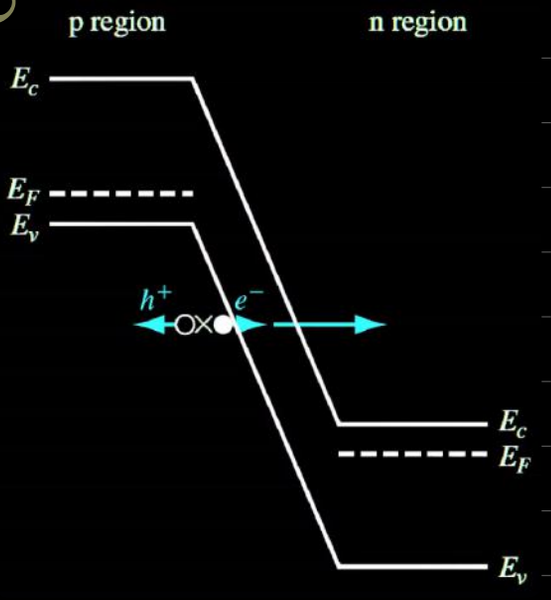
Breakdown Mechanisms In Diodes:



There are two types of breakdown mechanisms in reverse bias namely Zener breakdown which happens in heavily doped p-n junctions at relatively smaller voltages and Avalanche Breakdown, which happens in lightly doped p-n junctions at relatively higher voltages.

Zener Breakdown:



This type of breakdown occurs in heavily doped p-n junction. Heavily doped p-n junctions have very narrow depletion regions (of the order of 10^{-6} m or smaller). When a reverse voltage is applied to the diode practically, entire drop occurs across the depletion region causing very large electric fields. If the voltage is large enough it could create electric field of about 3×10^7 V/m across the narrow layer. This electric field is strong enough to break covalent bonds in the lattice thus generating electron hole pairs. This effect is called <u>Zener Effect</u>. Even a small further increase in reverse voltage produces a large number of charge carriers, thus junction has very low resistance after the breakdown. This process of emission of electrons due to strong electric field is called <u>internal field emission</u> or <u>field ionization</u> and requires field of the order of 10^6 to 10^7 V/m.

→ Typical zener breakdown voltages are less than 4V

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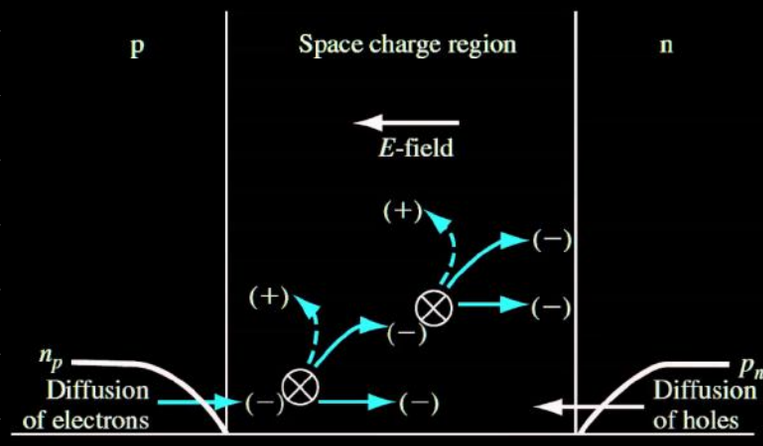
AVALANCHE BREAKDOWN:



This type of breakdown occurs in lightly doped p-n junctions which have relatively wider depletion regions and occurs at relatively higher voltages. Here electric field is not strong enough to break down the covalent bonds, however here thermally generated minority carriers gain sufficiently large kinetic energy while travelling through the depletion layer. They collide with semi-conductor atoms with high KE which leads to breaking down of covalent bonds and thus generating electron-hole pairs. The newly generated charge carriers are also accelerated by this field resulting in more collisions and further production of charge carriers. This is somewhat like a snow Avalanche where falling snow gathers more and more snow as it tumbles down the slopes of the hills and hence the name "Avalanche". The diodes operating in this mode are called Avalanche diodes. The generation of large number of charge carriers reduces the junction resistance drastically causing a sharp rise in current.

→ Typical Avalanche breakdown voltages are more than 6V. Between 4 and 6 Volts, both types of mechanisms can be present in the diode.

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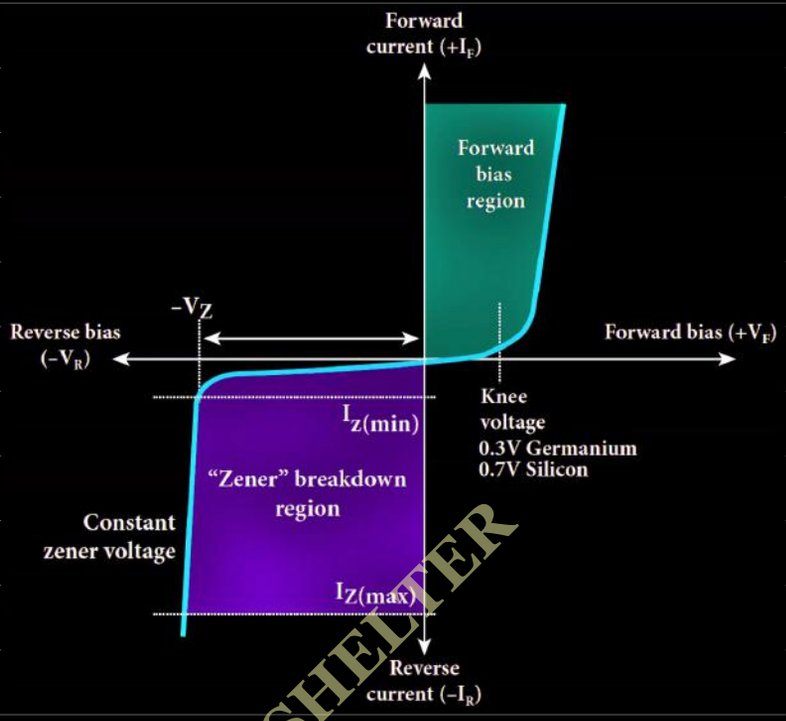
SPECIAL PURPOSE DIODES:

ZENER DIODE:

A zener diode is a heavily doped p-n junction used in reverse breakdown region. The circuit symbol for zener diode is slightly different from that of a normal diode (see figure).

Zener diode
(a) commercial picture (b) circuit symbol

Zener Diode Characteristics:



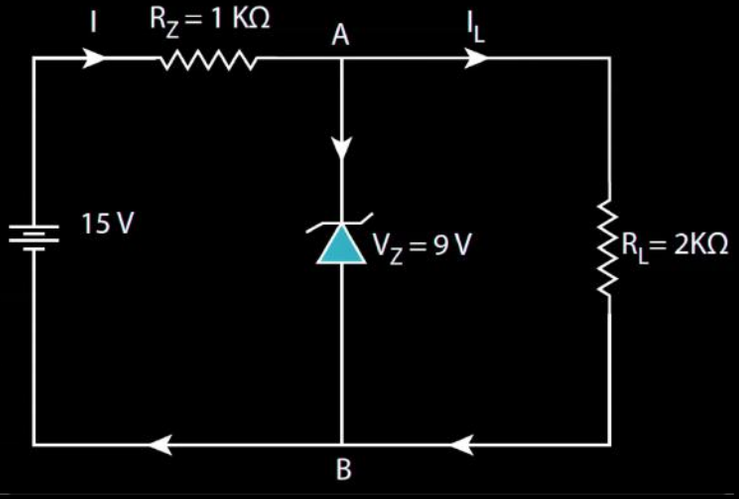
The V-I characteristics of a zener diode are similar to that of a normal diode except that here we can flow significantly larger current in reverse bias without damaging the diode (in usual diodes we will typically have currents in μA). There is a certain minimum current to sustain a reverse breakdown. Also for safe operation of zener diodes there is also a maximum current limit which is usually indicated on the diode. (Zener diodes are also marked with Power rating, which is the product of maximum allowable current and Zener voltage, which is nearly constant).

Uses Of Zener Diodes:

1. As voltage regulators
2. Calibrating voltages
3. Providing fix reference voltages in a network
4. Protection of Gadgets against accidental damage due to application of excessive voltage accidentally.

Example:

Find the current through the zener diode when the load resistance is 2kΩ.



Solution: $i_L = 9/2 = 4.5\,mA$.

$i = (15-9)/1 = 6\,mA$

Thus current through zener diode must be $6 - 4.5 = 1.5\,mA$ ans.

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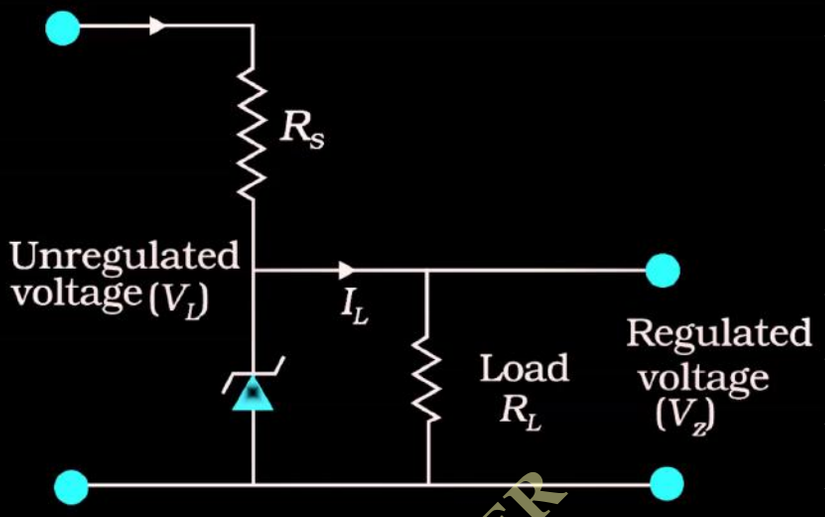
Zener Diode As A DC Voltage Regulator:

In reverse breakdown mode, a zener diode maintains a constant voltage irrespective of the current flow. Thus any device connected in parallel to a zener diode will also maintain a potential difference equal to zener breakdown voltage. If there are any voltage fluctuations in the unregulated power supply, zener diode will pass through it any excess current (see figure) thus protecting the load from any damage due to voltage fluctuations. To obtain a good regulation, typically R_S is so chosen that zener diode current is about 5 times the load current (as per NCERT).



Zener diode as DC voltage regulator

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NCERT Example:

Example 14.5 In a Zener regulated power supply a Zener diode with $V_Z = 6.0$ V is used for regulation. The load current is to be 4.0 mA and the unregulated input is 10.0 V. What should be the value of series resistor R_S?

Solution

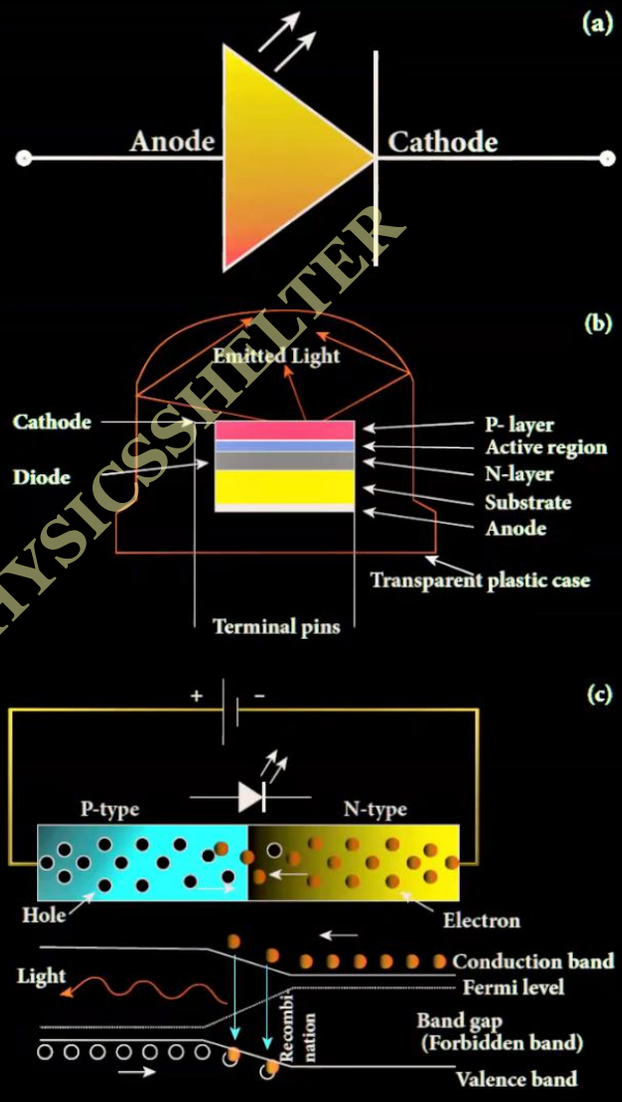
The value of R_S should be such that the current through the Zener diode is much larger than the load current. This is to have good load regulation. Choose Zener current as five times the load current, i.e., $I_Z = 20$ mA. The total current through R_S is, therefore, 24 mA. The voltage drop across R_S is $10.0 - 6.0 = 4.0$ V. This gives $R_S = 4.0\text{V}/(24 \times 10^{-3})\ \text{A} = 167\ \Omega$. The nearest value of carbon resistor is 150 Ω. So, a series resistor of 150 Ω is appropriate. Note that slight variation in the value of the resistor does not matter, what is important is that the current I_Z should be sufficiently larger than I_L.

Optoelectronic Devices:

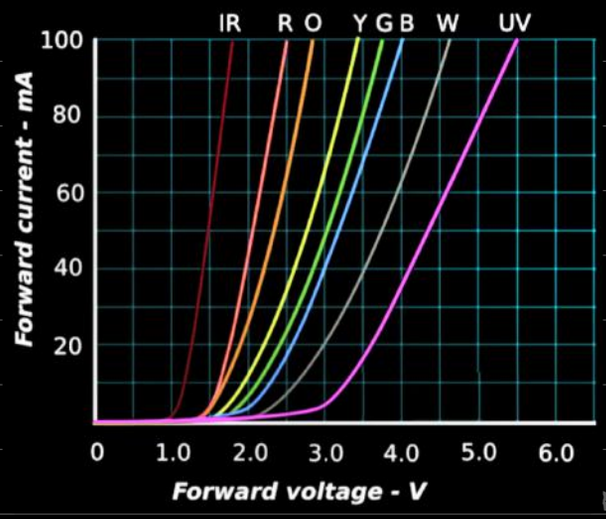
Opto-electronics deals with devices which convert electrical energy to light or vice-versa. We will be mainly discussing three types of optical devices namely, light emitting diodes (LEDs), Photo-diodes and solar cells.

Light Emitting Diode:

It is a heavily biased p-n junction which under forward bias emits radiation. Recall that forward bias involves minority carrier injection (i.e electrons injected in to p side and holes injected in to n side). Thus at the junction concentration of minority carriers increases as compared to the equilibrium. These minority carriers combine with majority carriers (electron-hole recombination) and in this process of recombination, energy is released in form of photons. The energy of photons is equal to or slightly less than the band gap. Thus according to the band gap of the material used, we can generate light of various colors. When the current is small, light intensity is small, the intensity increases with increase in forward current up to a point but then with further increase, the crystal heats up and emission efficiency decreases thus decreasing the intensity of light. LEDs are biased so as to maximize the light emitting efficiency. VI characteristics of LEDs are similar to that of a silicon diode



(a) Circuit symbol of LED
(b) Inside view of LED (c) Schematic diagram to explain recombination process



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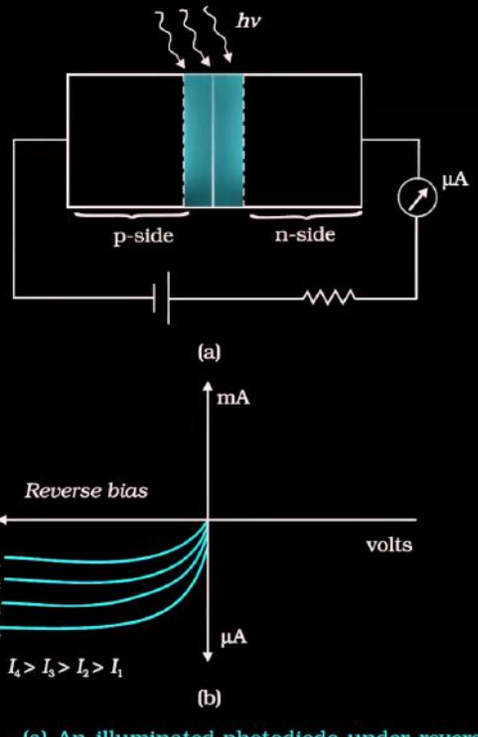
but the threshold (knee / cut-in) voltages are much higher and slightly different for each colour (see graph). Reverse breakdown voltages of LEDs are very low (~ 5V) so care should be taken to avoid such reverse voltages otherwise LED may get damaged. For visible light, the band gap of LED should be between 1.8 eV and 3.0 eV (corresponding to visible photons energy). LEDs are made from the compound Gallium Arsenide Phosphide ($GaAs_{1-x}P_x$). For instance, $GaAs_{0.6}P_{0.4}$ is used for red LED, GaAs ($E_g = 1.4$ eV) is used for infrared LED

Uses Of LEDs: Remote controls, Burglar alarm systems and optical communication etc.

Advantages Of LEDs:

1. Low operational voltage and less power consumption
2. Fast action and no warm up time required
3. Nearly monochromatic light (Band width about 100 - 500 Å)
4. Long life and ruggedness
5. Fast on-off switching ability

Photodiode: A photo diode has a transparent window to allow light to fall around its depletion layer. It is essentially a reverse biased p-n junction which detects optical signals. When light falls on the depletion layer, it generates e-h pairs. Due to presence of electric field in the depletion layer, electrons are quickly swept towards n side and holes towards p side thus generating an EMF. This EMF can be used to drive current through and external load.
The reverse bias is used because it makes it easier to observe small fluctuations in



(a) An illuminated photodiode under reverse bias, (b) I-V characteristics of a photodiode for different illumination intensity $I_4 > I_3 > I_2 > I_1$.

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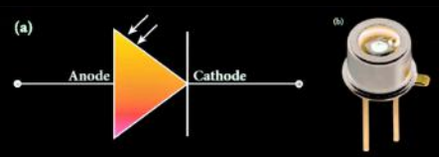
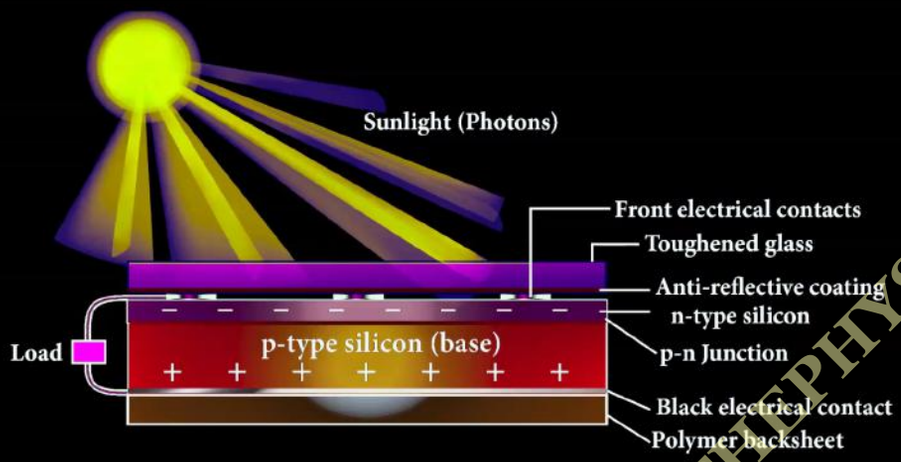
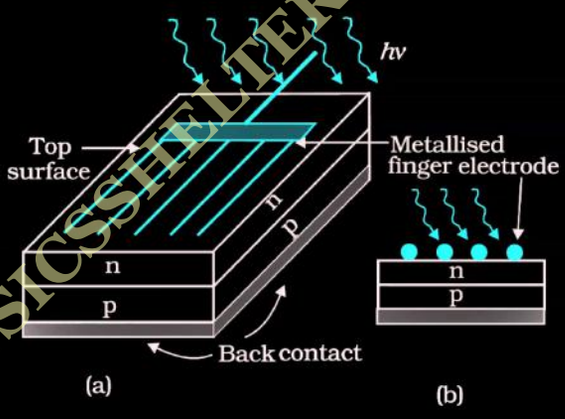
current with change in intensity of light. In forward bias, natural current itself is quite large making it difficult to perceive small changes in currents in response to light intensity variations. The VI characteristics, circuit, symbol and photograph of Photodiodes are shown in the figures above.



(a) Circuit symbol (b) Schematic view of photodiode

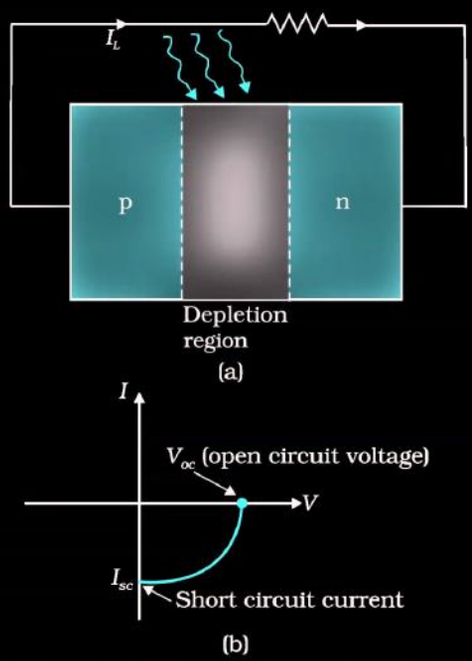
SOLAR CELL:





(a) Typical p-n junction solar cell
(b) Cross-sectional view.

A solar cell is an unbiased p-n junction in which EMF is generated when solar radiation falls on it. For making a solar cell we take a 300μm p type silicon wafer on which a very thin layer (0.3μm) of n type layer is grown. On the top of n type layer we have a finger electrode so as to allow light to pass through (it occupies less than 15% of top area). On the lower side the p type layer is coated with metal (back contact).



(a) A typicalilluminated p-n junction solar cell;
(b) *I-V* characteristics of a solar cell.

The generation of EMF by solar cell can be understood as follows:

① Due to light falling on depletion layer, e-h pairs are generated

② Due to strong electric field in depletion layer electrons are swept towards n side and holes towards p side (Gain KE).

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③ Electrons reaching n side are collected by front contact and holes are collected by back contact thus solar cell behaves like a battery with p side serving as positive terminal and n side serving as negative terminal of the battery.

When an external load is connected, there is some current through it accordingly terminal voltage across the solar cell varies due to potential drop across internal resistance of the cell. The VI graph of solar cell is shown in the figure above. Notice that the graph is not a straight line, rather a curve due to non-ohmic behavior of silicon. At open circuit, the voltage is maximum and at short circuit, the terminal voltage becomes zero.

Example 14.7 Why are Si and GaAs are preferred materials for solar cells?

Solution The solar radiation spectrum received by us is shown in Fig. 14.26.

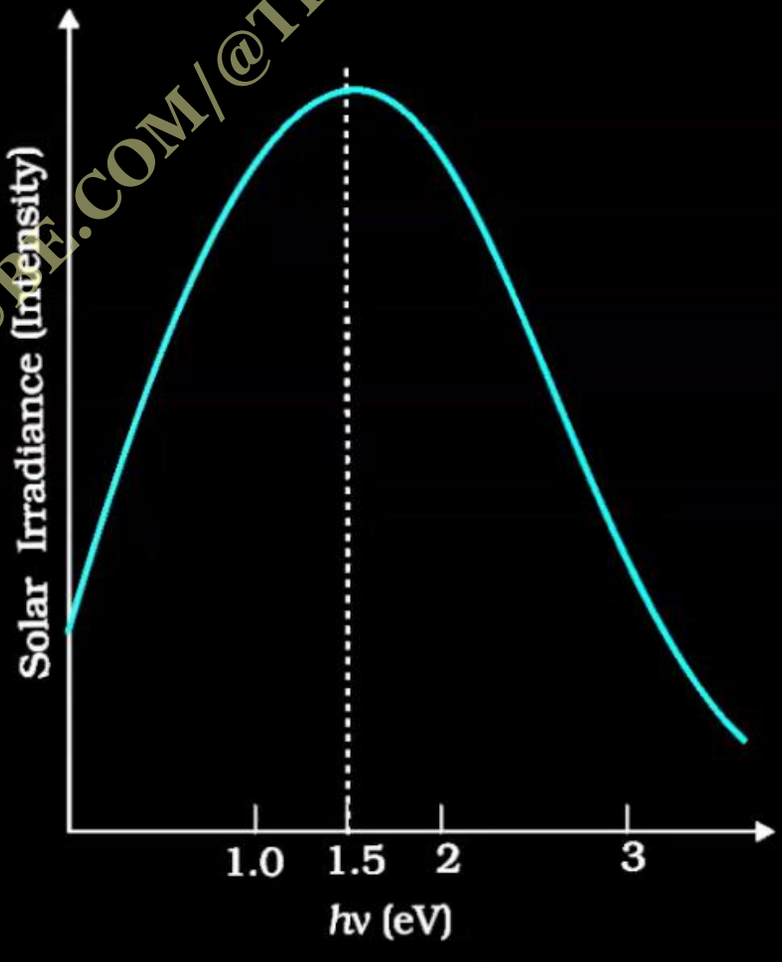

FIGURE 14.26

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The maxima is near 1.5 eV. For photo-excitation, $h\nu > E_g$. Hence, semiconductor with band gap ~1.5 eV or lower is likely to give better solar conversion efficiency. Silicon has $E_g \sim 1.1$ eV while for GaAs it is ~1.53 eV. In fact, GaAs is better (in spite of its higher band gap) than Si because of its relatively higher absorption coefficient. If we choose materials like CdS or CdSe ($E_g \sim 2.4$ eV), we can use only the high energy component of the solar energy for photo-conversion and a significant part of energy will be of no use.

The question arises: why we do not use material like PbS ($E_g \sim 0.4$ eV) which satisfy the condition $h\nu > E_g$ for ν maxima corresponding to the solar radiation spectra? If we do so, most of the solar radiation will be absorbed on the *top-layer* of solar cell and will not reach in or near the depletion region. For effective electron-hole separation, due to the junction field, we want the photo-generation to occur in the junction region only.

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